

spark_temperature_eda

November 30, 2025

1 Global Temperature Analysis

1.1 Analysis of Temperature with country and major city

1.1.1 By Utsav Chatterjee

```
[1]: # === Spark Setup ===  
  
import findspark  
findspark.init()  
  
from pyspark.sql import SparkSession  
  
spark = (  
    SparkSession.builder  
        .appName("GlobalTemperatureAnalysis")  
        .master("local[*]")  
        .config("spark.driver.memory", "4g")  
        .config("spark.executor.memory", "4g")  
        .getOrCreate()  
)  
  
spark
```

```
25/11/30 12:56:09 WARN Utils: Your hostname, UTSAV-LAPTOP resolves to a loopback  
address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)  
25/11/30 12:56:09 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another  
address  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use  
setLogLevel(newLevel).  
25/11/30 12:56:10 WARN NativeCodeLoader: Unable to load native-hadoop library  
for your platform... using builtin-java classes where applicable  
25/11/30 12:56:10 WARN Utils: Service 'SparkUI' could not bind on port 4040.  
Attempting port 4041.
```

```
[1]: <pyspark.sql.session.SparkSession at 0x7f18c81baf60>
```

1.2 Load the CSVs

```
[5]: df_global = spark.read.csv("GlobalTemperatures.csv", header=True,
    ↪inferSchema=True)
df_country = spark.read.csv("GlobalLandTemperaturesByCountry.csv", header=True,
    ↪inferSchema=True)
df_city = spark.read.csv("GlobalLandTemperaturesByMajorCity.csv", header=True,
    ↪inferSchema=True)
```

```
[6]: df_global.printSchema()
df_country.printSchema()
df_city.printSchema()
```

```
root
|-- dt: date (nullable = true)
|-- LandAverageTemperature: double (nullable = true)
|-- LandAverageTemperatureUncertainty: double (nullable = true)
|-- LandMaxTemperature: double (nullable = true)
|-- LandMaxTemperatureUncertainty: double (nullable = true)
|-- LandMinTemperature: double (nullable = true)
|-- LandMinTemperatureUncertainty: double (nullable = true)
|-- LandAndOceanAverageTemperature: double (nullable = true)
|-- LandAndOceanAverageTemperatureUncertainty: double (nullable = true)
```

```
root
|-- dt: date (nullable = true)
|-- AverageTemperature: double (nullable = true)
|-- AverageTemperatureUncertainty: double (nullable = true)
|-- Country: string (nullable = true)
```

```
root
|-- dt: date (nullable = true)
|-- AverageTemperature: double (nullable = true)
|-- AverageTemperatureUncertainty: double (nullable = true)
|-- City: string (nullable = true)
|-- Country: string (nullable = true)
|-- Latitude: string (nullable = true)
|-- Longitude: string (nullable = true)
```

```
[7]: df_global.show(5)
```

```
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      dt|LandAverageTemperature|LandAverageTemperatureUncertainty|LandMaxTemp
```

```

erature|LandMaxTemperatureUncertainty|LandMinTemperature|LandMinTemperatureUncer
tainty|LandAndOceanAverageTemperature|LandAndOceanAverageTemperatureUncertainty|
+-----+-----+-----+-----+
-----+-----+-----+-----+
+-----+-----+-----+-----+
|1750-01-01| 3.0340000000000003| | 3.574|
NULL| NULL| NULL|
NULL| NULL| NULL| NULL|
|1750-02-01| 3.083| | 3.702|
NULL| NULL| NULL|
NULL| NULL| NULL| NULL|
|1750-03-01| 5.626| | 3.076|
NULL| NULL| NULL|
NULL| NULL| NULL| NULL|
|1750-04-01| 8.49| | 2.451|
NULL| NULL| NULL|
NULL| NULL| NULL| NULL|
|1750-05-01| 11.573| | 2.072|
NULL| NULL| NULL|
NULL| NULL| NULL| NULL|
+-----+-----+-----+-----+
-----+-----+-----+-----+
-----+-----+-----+-----+

```

only showing top 5 rows

```
[8]: df_country.show(5)
```

```

+-----+-----+-----+-----+
| dt|AverageTemperature|AverageTemperatureUncertainty|Country|
+-----+-----+-----+-----+
|1743-11-01|4.3839999999999995| 2.294| Åland|
|1743-12-01| NULL| NULL| Åland|
|1744-01-01| NULL| NULL| Åland|
|1744-02-01| NULL| NULL| Åland|
|1744-03-01| NULL| NULL| Åland|
+-----+-----+-----+-----+

```

only showing top 5 rows

```
[9]: df_city.show(5)
```

```

+-----+-----+-----+-----+
--+-----+-----+
| dt|AverageTemperature|AverageTemperatureUncertainty| City|
Country|Latitude|Longitude|
+-----+-----+-----+-----+
--+-----+-----+

```

```

|1849-01-01|          26.704|          1.435|Abidjan|Côte
D'Ivoire|   5.63N|   3.23W|
|1849-02-01|          27.434|          1.362|Abidjan|Côte
D'Ivoire|   5.63N|   3.23W|
|1849-03-01|          28.101|          1.612|Abidjan|Côte
D'Ivoire|   5.63N|   3.23W|
|1849-04-01|          26.14|          1.3869999999999998|Abidjan|Côte
D'Ivoire|   5.63N|   3.23W|
|1849-05-01|          25.427|          1.2|Abidjan|Côte
D'Ivoire|   5.63N|   3.23W|
+-----+-----+-----+-----+-----+
--+-----+-----+
only showing top 5 rows

```

1.3 Checking Missing Values for each

```

[10]: # Missing value count for df_global
from pyspark.sql.functions import col, sum as spark_sum, when

missing_global = df_global.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in df_global.columns
])

missing_global.show(truncate=False)

+---+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|dt |LandAverageTemperature|LandAverageTemperatureUncertainty|LandMaxTemperature|
|LandMaxTemperatureUncertainty|LandMinTemperature|LandMinTemperatureUncertainty|
|LandAndOceanAverageTemperature|LandAndOceanAverageTemperatureUncertainty|
+---+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|0  |12          |12          |1200          |
|1200          |1200          |1200          |
|1200          |1200          |              |
+---+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+

```

```

[11]: df_global.describe().show()

```

25/11/27 00:57:50 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting

```
'spark.sql.debug.maxToStringFields'.
```

```
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|summary|LandAverageTemperature|LandAverageTemperatureUncertainty|LandMaxTempera
ture|LandMaxTemperatureUncertainty|LandMinTemperature|LandMinTemperatureUncertai
nty|LandAndOceanAverageTemperature|LandAndOceanAverageTemperatureUncertainty|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|  count|                3180|                3180|
1992|                1992|                1992|
1992|                1992|                1992|
|  mean|    8.374731132075466|
0.9384679245283027|14.350600903614467|
0.47978162650602346|2.7435953815261076|    0.431848895582329|
15.212565763052215|    0.12853212851405638|
| stddev|    4.3813097707251805|    1.0964397949200564|
4.309578965948537|    0.5832029574744946| 4.155835320049621|
0.4458378370608339|    1.2740929544373867|
0.07358679600543891|
|  min|    -2.08|    0.034|
5.9|    0.044|    -5.407|
0.045|    12.475|    0.042|
|  max|    19.021|    7.88|
21.32|    4.373|    9.715|
3.498|    17.611|    0.457|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+-----+-----+
```

```
[12]: # Missing Values in df_city
missing_city = df_city.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in df_city.columns
])

missing_city.show(truncate=False)
```

```
+---+-----+-----+-----+-----+-----+
+-----+
|dt |AverageTemperature|AverageTemperatureUncertainty|City|Country|Latitude|Long
itude|
+---+-----+-----+-----+-----+-----+
+-----+
|0  |11002                |11002                |0  |0  |0  |0  |
```

```
|
+---+-----+-----+-----+-----+-----+
-----+
```

```
[13]: df_city.describe().show()
```

```
[Stage 18:=====> (2 + 2) / 4]
+---+-----+-----+-----+-----+-----+
-----+
|summary|AverageTemperature|AverageTemperatureUncertainty|City|
Country|Latitude|Longitude|
+---+-----+-----+-----+-----+-----+
-----+
|count|228175|228175|239177|239177|
239177|239177|
|mean|18.125968852854175|0.9693434381505347|NULL|NULL|
NULL|NULL|
|stddev|10.02479971048484|0.979643911437092|NULL|NULL|
NULL|NULL|
|min|-26.772|0.04|Abidjan|Afghanistan|
0.80N|0.00W|
|max|38.283|14.037|Xian|Zimbabwe|
8.84S|99.91E|
+---+-----+-----+-----+-----+-----+
-----+
```

```
[14]: # Missing Values in df_country
missing_country = df_country.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in df_country.columns
])

missing_country.show(truncate=False)
```

```
+---+-----+-----+-----+
|dt |AverageTemperature|AverageTemperatureUncertainty|Country|
+---+-----+-----+-----+
|0 |32651|31912|0 |
+---+-----+-----+-----+
```

```
[15]: df_country.describe().show()
```

```
[Stage 24:=====> (1 + 3) / 4]
```

summary	AverageTemperature	AverageTemperatureUncertainty	Country
count	544811	545550	577462
mean	17.19335423293594	1.0190569003757632	NULL
stddev	10.953966445121155	1.201930386633437	NULL
min	-37.658	0.052000000000000005	Afghanistan
max	38.84200000000001	15.003	Åland

1.4 Data Quality Assessment and Statistical Insights

In this section, we evaluate the three datasets used in the project: - **Global Temperatures** - **City-Level Temperatures** - **Country-Level Temperatures**

For each dataset, we examined:

1. Missing values
2. Descriptive statistics
3. Data quality observations
4. Implications for the next steps in preprocessing

1.5 1. Global Temperature Dataset (df_global)

1.5.1 Missing Values

The global dataset shows missing values in multiple fields including: - **LandAverageTemperature** - **LandAverageTemperatureUncertainty** - **LandMaxTemperature** - **LandMinTemperature** - **LandAndOceanAverageTemperature** - Corresponding uncertainty columns

The presence of nulls here is expected due to incomplete climate recordings in earlier historical periods.

1.5.2 Descriptive Statistics Insights

- Contains **3192 total rows**, spanning **years 1750–2015**.
- **LandAverageTemperature** ranges approximately from **−2.08°C to 19.02°C**.
- Uncertainty values range from **0.03 to ~7.8**, indicating older records have higher measurement error.
- Seasonal and Period entries exist, but descriptive stats show them as NULL because they are categorical.
- Temperature variability across seasons and centuries is evident from the standard deviations.

1.5.3 Interpretation

The dataset is generally clean but contains:

- Gaps in certain time periods
- Higher uncertainty in early years
- Multiple temperature measurement types

The dataset will be treated as a **standalone global trend dataset** and **not merged** with city/country data.

1.6 2. City Temperature Dataset (df_city)

1.6.1 Missing Values

The city dataset contains: - Missing temperature and uncertainty values. - Some missing or invalid City and Country entries (shown as 0 in Spark results). - Complete coverage for Latitude and Longitude.

1.6.2 Descriptive Statistics Insights

- Contains **239,177 rows** covering years **1743–2013**.
- Temperature ranges from **−26.77°C to 38.28°C**, showing large climatic diversity.
- Uncertainty ranges up to **14.0**, indicating measurement noise in historical data.
- Seasonal and Period columns appear as NULL in descriptive stats because they are categorical.
- Latitude/Longitude appear as strings (e.g., “34.56N”), meaning they must be kept as categorical text fields.

1.6.3 Interpretation

This dataset has: - Good temporal depth
- Accurate city-level granularity
- Slight missingness that will require imputation
- All the fields required for merging with country data later

1.7 3. Country Temperature Dataset (df_country)

1.7.1 Missing Values

The country dataset also contains missing values in: - **AverageTemperature** - **AverageTemperatureUncertainty** - Some Country entries appear invalid (0)

However, the presence of valid country names (min = “Afghanistan”, max = “Åland”) suggests that these anomalies are limited.

1.7.2 Descriptive Statistics Insights

- Contains **577,462 rows**, the largest dataset among the three.
- Covers years **1743–2013**, same as city data.
- Temperature ranges from **−37.65°C to 38.84°C**.
- Uncertainty ranges up to **15**, consistent with historical records.
- Mean temperature is around **17.19°C**.
- Strong seasonality and century-level variability are expected.

1.7.3 Interpretation

This dataset serves as the **country-level aggregate** and aligns perfectly with the city dataset for a future merge. Missing values require handling but are not structurally problematic.

1.8 Next Steps

Based on the missing-value analysis and statistical summaries:

1. **Handle Missing Values**
 - Impute missing temperature fields using dataset-wise mean or median.
 - Preserve categorical fields (City, Country, Season, Period) as-is.
2. **Feature Engineering**
 - Extract Year, Month, Season, Period cleanly for all datasets (already added to Spark datasets).
3. **Dataset Alignment Before Merge**
 - Only the **City and Country datasets** will be merged (global dataset remains separate for global trend analysis).
4. **Consistency Checks**
 - Validate matching dt + Country + City keys before merging.
 - Remove invalid entries (0 in Country/City columns).
5. **Proceed to EDA**
 - Year-wise trends
 - Seasonal variation
 - Country vs City comparisons
 - Distribution plots
 - Temperature anomalies

After establishing quality and consistency, we can begin preparing the **final merged dataset** for modeling and visualization (Power BI).

1.9 Cleaning & Feature Engineering

```
[16]: # Add Year and Month Columns
      from pyspark.sql.functions import year, month, col

      # Global dataset
      df_global = df_global.withColumn("Year", year(col("dt"))) \
                          .withColumn("Month", month(col("dt")))
```

```
# Country dataset
df_country = df_country.withColumn("Year", year(col("dt"))) \
                        .withColumn("Month", month(col("dt")))

# City dataset
df_city = df_city.withColumn("Year", year(col("dt"))) \
                .withColumn("Month", month(col("dt")))
```

1.9.1 Step 1 — Extract Year and Month

We extract the Year and Month components from the dt (date) column for all three datasets. This supports seasonal analysis, temporal aggregations, and merging operations later on.

Spark's year() and month() functions ensure efficient computation on large datasets.

```
[17]: # Create Season Column for All Three Datasets
from pyspark.sql.functions import when

def add_season(df):
    return df.withColumn(
        "Season",
        when(col("Month").isin(12, 1, 2), "Winter")
        .when(col("Month").isin(3, 4, 5), "Spring")
        .when(col("Month").isin(6, 7, 8), "Summer")
        .otherwise("Autumn")
    )

df_global = add_season(df_global)
df_country = add_season(df_country)
df_city = add_season(df_city)
```

1.9.2 Step 2 — Create Season Feature

We classify each row into meteorological seasons based on the Month value:

- **Winter:** December–February
- **Spring:** March–May
- **Summer:** June–August
- **Autumn:** September–November

This feature allows seasonal trend analysis, temperature variability studies, and supports group-by operations during EDA.

```
[18]: # Add Period Column (Historical Time Segments) 50 years
def add_period(df):
    return df.withColumn(
```

```

        "Period",
        when(col("Year") < 1800, "Pre-1800")
        .when((col("Year") >= 1800) & (col("Year") < 1850), "1800-1849")
        .when((col("Year") >= 1850) & (col("Year") < 1900), "1850-1899")
        .when((col("Year") >= 1900) & (col("Year") < 1950), "1900-1949")
        .when((col("Year") >= 1950) & (col("Year") < 2000), "1950-1999")
        .otherwise("2000+")
    )

df_global = add_period(df_global)
df_country = add_period(df_country)
df_city = add_period(df_city)

```

1.9.3 Step 3 — Create Period Feature

We segment the timeline into historical periods to analyze long-term temperature trends across centuries. These ranges are divided into 50 years periods:

- Pre-1800
- 1800–1849
- 1850–1899
- 1900–1949
- 1950–1999
- 2000+

This feature helps identify century-level climate patterns and supports trend visualization.

```

[19]: # Derive average-temperature columns from min/max where possible
from pyspark.sql.functions import col, to_date, year, month, when, coalesce
from pyspark.sql.functions import expr

df_global = df_global.withColumn(
    "Global_AvgTemp_derived",
    coalesce(
        col("LandAverageTemperature"),
        ((col("LandMaxTemperature") + col("LandMinTemperature"))/2)
    )
)

[20]: df_country = df_country.withColumnRenamed(
    "AverageTemperature", "AverageTemperature_country"
).withColumnRenamed(
    "AverageTemperatureUncertainty", "AverageTemperatureUncertainty_country"

```

```
)
```

```
[21]: df_city = df_city.withColumnRenamed(
      "AverageTemperature", "AverageTemperature_city"
    ).withColumnRenamed(
      "AverageTemperatureUncertainty", "AverageTemperatureUncertainty_city"
    )
```

```
[22]: df_country = df_country.drop("Season", "Period", "Year", "Month")
```

```
[23]: # Merge City & Country
merged_df = df_city.join(
    df_country,
    on=["dt", "Country"],
    how="inner"
)
```

```
[24]: final_df = merged_df.select(
    col("dt").alias("Date"),
    col("AverageTemperature_country"),
    col("AverageTemperatureUncertainty_country"),
    col("Country"),
    col("AverageTemperature_city"),
    col("AverageTemperatureUncertainty_city"),
    col("City"),
    col("Latitude"),
    col("Longitude"),
    col("Season"),
    col("Period")
)
```

```
[25]: final_df.show(20, truncate=False)
```

```
[Stage 28:=====> (2 + 2) / 4]
```

Date	AverageTemperature_country	AverageTemperatureUncertainty_country	Country	AverageTemperature_city	AverageTemperatureUncertainty_city	City	Latitude	Longitude	Season	Period
1743-11-01	10.203	2.03	France	7.648	1.749	Paris	49.03N	2.45E	Autumn	Pre-1800

1743-11-01 5.468	1.832	
Germany	6.3260000000000005	1.601
Berlin 52.24N	13.14E	Autumn Pre-1800
1743-11-01 9.538	1.944	
Italy	8.2480000000000001	1.917
42.59N	13.09E	Autumn Pre-1800
1743-11-01 1.898	2.8080000000000003	
Ukraine	0.895	2.092
50.63N	31.69E	Autumn Pre-1800
1743-12-01 NULL	NULL	
France	NULL	NULL
49.03N	2.45E	Winter Pre-1800
1743-12-01 NULL	NULL	
Ukraine	NULL	NULL
50.63N	31.69E	Winter Pre-1800
1744-01-01 NULL	NULL	
Ukraine	NULL	NULL
50.63N	31.69E	Winter Pre-1800
1744-01-01 NULL	NULL	
United Kingdom NULL	NULL	
London 52.24N	0.00W	Winter Pre-1800
1744-02-01 NULL	NULL	
France	NULL	NULL
49.03N	2.45E	Winter Pre-1800
1744-02-01 NULL	NULL	
Spain	NULL	NULL
Madrid 40.99N	4.26W	Winter Pre-1800
1744-02-01 NULL	NULL	
United Kingdom NULL	NULL	
London 52.24N	0.00W	Winter Pre-1800
1744-03-01 NULL	NULL	
Spain	NULL	NULL
Madrid 40.99N	4.26W	Spring Pre-1800
1744-03-01 NULL	NULL	
United Kingdom NULL	NULL	
London 52.24N	0.00W	Spring Pre-1800
1744-04-01 8.437999999999999	2.7460000000000004	
Germany	9.536	2.761
Berlin 52.24N	13.14E	Spring Pre-1800
1744-04-01 13.567	2.325	
Spain	11.189	2.17
Madrid 40.99N	4.26W	Spring Pre-1800
1744-04-01 7.619	2.513	
United Kingdom 8.296	2.501	
London 52.24N	0.00W	Spring Pre-1800
1744-06-01 14.262	1.639	
Germany	14.8090000000000001	1.398
Berlin 52.24N	13.14E	Summer Pre-1800

```

|1744-07-01|21.131999999999998      |1.904
|Italy      |21.03                        |1.71                        |Rome
|42.59N   |13.09E   |Summer|Pre-1800|
|1744-07-01|19.673                      |2.7
|Ukraine   |19.3890000000000006          |1.592                      |Kiev
|50.63N   |31.69E   |Summer|Pre-1800|
|1744-08-01|NULL                          |NULL
|France     |NULL                          |NULL                        |Paris
|49.03N   |2.45E     |Summer|Pre-1800|
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+
only showing top 20 rows

```

1.9.4 Fixing Column Names Before Merge

The merge failed because the renamed columns `AverageTemperature_country` and `AverageTemperature_city` did not exist prior to joining.

We explicitly rename the temperature and uncertainty columns in the country and city dataframes. We also drop duplicate engineered columns from country before merging. After this correction, the merge and select operations work exactly as expected.

```

[26]: # Missing Values for final_df
from pyspark.sql.functions import col, sum as spark_sum, when

missing_final = final_df.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in final_df.columns
])

missing_final.show(truncate=False)

```

```

[Stage 33:=====>                                     (2 + 2) / 4]
+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
|Date|AverageTemperature_country|AverageTemperatureUncertainty_country|Country|A
verageTemperature_city|AverageTemperatureUncertainty_city|City|Latitude|Longitud
e|Season|Period|
+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
|0   |10950                      |10950                      |0      |0
|9745                      |9745                      |0      |0      |0
|0   |0      |

```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+

```

```
[27]: final_df.describe().show(truncate=False)
```

```

[Stage 45:> (0 + 4) / 4]
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|summary|AverageTemperature_country|AverageTemperatureUncertainty_country|Countr
y      |AverageTemperature_city|AverageTemperatureUncertainty_city|City
|Latitude|Longitude|Season|Period  |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|count  |222556                |222556                |233506
|223761                |223761                |233506 |233506
|233506 |233506|233506 |
|mean   |17.04014827279453    |0.7963963811355309   |NULL
|18.26369776234467    |0.9300551570649113   |NULL   |NULL
|NULL    |NULL |NULL    |
|stddev  |11.051996933111425   |0.8630554499363159   |NULL
|9.951802135657122    |0.9261036904403022   |NULL   |NULL
|NULL    |NULL |NULL    |
|min     |-30.577              |0.053
|Afghanistan|-26.772              |0.04
|Abidjan|0.80N |0.00W |Autumn|1800-1849|
|max     |37.400999999999996   |13.954
|Zimbabwe |38.283              |14.037
|8.84S   |99.91E |Winter|Pre-1800 |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+

```

```

[28]: # To Handle the remaining missing values
# Compute 50-year × Season × Country averages
from pyspark.sql.functions import avg

impute_table = final_df.groupBy(
    "Country", "Period", "Season"
).agg(

```

```

    avg("AverageTemperature_country").alias("mean_country_temp"),
    avg("AverageTemperature_city").alias("mean_city_temp"),
    avg("AverageTemperatureUncertainty_country").alias("mean_country_unc"),
    avg("AverageTemperatureUncertainty_city").alias("mean_city_unc")
)

```

1.9.5 Step — Create Imputation Table (Country × Period × Season)

We calculate seasonal temperature averages for each 50-year period and country pair. The missing values were replaced using the corresponding period/season aggregates.

This produces a lookup table used for precise, context-aware imputation.

```

[29]: # Final Join
final_df2 = final_df.join(
    impute_table,
    on=["Country", "Period", "Season"],
    how="left"
)

```

1.9.6 Attach Seasonal Period Averages to Each Row

We perform a left join between the full dataset and the imputation table, so every row now includes the corresponding seasonal and period averages.

This enables precise replacement of missing values without using global means.

```

[30]: # Fill Missing Values Using the Computed Averages
from pyspark.sql.functions import coalesce

final_df3 = final_df2.withColumn(
    "AverageTemperature_country_filled",
    coalesce(col("AverageTemperature_country"), col("mean_country_temp"))
).withColumn(
    "AverageTemperature_city_filled",
    coalesce(col("AverageTemperature_city"), col("mean_city_temp"))
).withColumn(
    "AverageTemperatureUncertainty_country_filled",
    coalesce(col("AverageTemperatureUncertainty_country"),
    ↪col("mean_country_unc"))
).withColumn(
    "AverageTemperatureUncertainty_city_filled",
    coalesce(col("AverageTemperatureUncertainty_city"), col("mean_city_unc"))
)

```

1.9.7 Step — Impute Missing Values Using Seasonal × Period Averages

We replace missing temperature and uncertainty values using the aggregated Country × Period × Season averages. This replicates the logic from the original Pandas EDA and significantly reduces

missing values.

```
[31]: from pyspark.sql.functions import col, when, sum as spark_sum
```

```
final_df3.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in final_df3.columns
]).show(truncate=False)
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
|Country|Period|Season|Date|AverageTemperature_country|AverageTemperatureUncerta
inty_country|AverageTemperature_city|AverageTemperatureUncertainty_city|City|Lat
itude|Longitude|mean_country_temp|mean_city_temp|mean_country_unc|mean_city_unc|
AverageTemperature_country_filled|AverageTemperature_city_filled|AverageTemperat
ureUncertainty_country_filled|AverageTemperatureUncertainty_city_filled|
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
|0      |0      |0      |0      |10950      |10950      |0      |0      |0
|9745    |9745    |0      |0      |0      |0      |0
|0      |0      |0      |0      |0      |0      |0
|0      |0      |0      |0      |0      |0      |0
|
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
```

```
[32]: # List distinct country names in both datasets
```

```
countries_city = df_city.select("Country").distinct()
countries_country = df_country.select("Country").distinct()
```

```
city_names = [row["Country"] for row in countries_city.collect()]
country_names = [row["Country"] for row in countries_country.collect()]
```

```
# Print some samples
```

```
print("Sample city countries:", city_names[:20])
```

```
print("Sample country countries:", country_names[:20])
```

Sample city countries: ['Senegal', 'Turkey', 'Iraq', 'Germany', 'India', 'China', 'United States', 'Bangladesh', 'Thailand', 'Morocco', 'Côte D'Ivoire', 'Syria', 'Ethiopia', 'Brazil', 'Tanzania', 'Egypt', 'South Africa', 'Colombia', 'Pakistan', 'Russia']

Sample country countries: ['Chad', 'Anguilla', 'Burma', 'Eritrea', 'Djibouti', 'Åland', 'Europe', 'Comoros', 'Afghanistan', 'Cambodia', 'Africa', 'British Virgin Islands', 'Dominica', 'Algeria', 'Equatorial Guinea', 'Argentina', 'Angola', 'Belgium', 'Baker Island', 'Ecuador']

1.9.8 Country Name Mismatch Investigation

The merged dataset contained only invalid rows (Country = “0”), which indicates that the merge keys did not match. This usually happens when the city and country datasets contain country names in different formats—for example:

- “United States” vs “United States of America”
- “Côte d'Ivoire” vs “Cote d'Ivoire”
- “Korea” vs “South Korea”

To resolve this, we extract and compare distinct country names from both datasets. Based on this comparison, we will apply the same normalization logic used in the original Pandas notebook before merging.

```
[33]: city_countries = [row["Country"] for row in df_city.select("Country").
      ↪distinct().orderBy("Country").collect()]
      country_countries = [row["Country"] for row in df_country.select("Country").
      ↪distinct().orderBy("Country").collect()]

      print("CITY dataset countries:")
      for c in city_countries:
          print(c)

      print("\nCOUNTRY dataset countries:")
      for c in country_countries:
          print(c)
```

```
CITY dataset countries:
Afghanistan
Angola
Australia
Bangladesh
Brazil
Burma
Canada
Chile
China
Colombia
Congo (Democratic Republic Of The)
Côte D'Ivoire
```

Dominican Republic
Egypt
Ethiopia
France
Germany
India
Indonesia
Iran
Iraq
Italy
Japan
Kenya
Mexico
Morocco
Nigeria
Pakistan
Peru
Philippines
Russia
Saudi Arabia
Senegal
Singapore
Somalia
South Africa
South Korea
Spain
Sudan
Syria
Taiwan
Tanzania
Thailand
Turkey
Ukraine
United Kingdom
United States
Vietnam
Zimbabwe

COUNTRY dataset countries:

Afghanistan
Africa
Albania
Algeria
American Samoa
Andorra
Angola
Anguilla
Antarctica

Antigua And Barbuda
Argentina
Armenia
Aruba
Asia
Australia
Austria
Azerbaijan
Bahamas
Bahrain
Baker Island
Bangladesh
Barbados
Belarus
Belgium
Belize
Benin
Bhutan
Bolivia
Bonaire, Saint Eustatius And Saba
Bosnia And Herzegovina
Botswana
Brazil
British Virgin Islands
Bulgaria
Burkina Faso
Burma
Burundi
Cambodia
Cameroon
Canada
Cape Verde
Cayman Islands
Central African Republic
Chad
Chile
China
Christmas Island
Colombia
Comoros
Congo
Congo (Democratic Republic Of The)
Costa Rica
Croatia
Cuba
Curaçao
Cyprus
Czech Republic

Côte D'Ivoire
Denmark
Denmark (Europe)
Djibouti
Dominica
Dominican Republic
Ecuador
Egypt
El Salvador
Equatorial Guinea
Eritrea
Estonia
Ethiopia
Europe
Falkland Islands (Islas Malvinas)
Faroe Islands
Federated States Of Micronesia
Fiji
Finland
France
France (Europe)
French Guiana
French Polynesia
French Southern And Antarctic Lands
Gabon
Gambia
Gaza Strip
Georgia
Germany
Ghana
Greece
Greenland
Grenada
Guadeloupe
Guam
Guatemala
Guernsey
Guinea
Guinea Bissau
Guyana
Haiti
Heard Island And Mcdonald Islands
Honduras
Hong Kong
Hungary
Iceland
India
Indonesia

Iran
Iraq
Ireland
Isle Of Man
Israel
Italy
Jamaica
Japan
Jersey
Jordan
Kazakhstan
Kenya
Kingman Reef
Kiribati
Kuwait
Kyrgyzstan
Laos
Latvia
Lebanon
Lesotho
Liberia
Libya
Liechtenstein
Lithuania
Luxembourg
Macau
Macedonia
Madagascar
Malawi
Malaysia
Mali
Malta
Martinique
Mauritania
Mauritius
Mayotte
Mexico
Moldova
Monaco
Mongolia
Montenegro
Montserrat
Morocco
Mozambique
Namibia
Nepal
Netherlands
Netherlands (Europe)

New Caledonia
New Zealand
Nicaragua
Niger
Nigeria
Niue
North America
North Korea
Northern Mariana Islands
Norway
Oceania
Oman
Pakistan
Palau
Palestina
Palmyra Atoll
Panama
Papua New Guinea
Paraguay
Peru
Philippines
Poland
Portugal
Puerto Rico
Qatar
Reunion
Romania
Russia
Rwanda
Saint Barthélemy
Saint Kitts And Nevis
Saint Lucia
Saint Martin
Saint Pierre And Miquelon
Saint Vincent And The Grenadines
Samoa
San Marino
Sao Tome And Principe
Saudi Arabia
Senegal
Serbia
Seychelles
Sierra Leone
Singapore
Sint Maarten
Slovakia
Slovenia
Solomon Islands

Somalia
South Africa
South America
South Georgia And The South Sandwich Isla
South Korea
Spain
Sri Lanka
Sudan
Suriname
Svalbard And Jan Mayen
Swaziland
Sweden
Switzerland
Syria
Taiwan
Tajikistan
Tanzania
Thailand
Timor Leste
Togo
Tonga
Trinidad And Tobago
Tunisia
Turkey
Turkmenistan
Turks And Caicas Islands
Uganda
Ukraine
United Arab Emirates
United Kingdom
United Kingdom (Europe)
United States
Uruguay
Uzbekistan
Venezuela
Vietnam
Virgin Islands
Western Sahara
Yemen
Zambia
Zimbabwe
Åland

1.9.9 Step — Extract Full Unique Country Lists

We extract complete, sorted lists of unique country names from both the city and country datasets. This will allow us to construct a normalization mapping to align the names across datasets, enabling a correct merge on Country + Date.


```
[34]: #Filter COUNTRY dataset to ONLY include countries present in CITY dataset
city_country_list = [
"Afghanistan","Angola","Australia","Bangladesh","Brazil","Burma","Canada","Chile","China","Col
"Congo (Democratic Republic Of The)","Côte D'Ivoire","Dominican_
↳Republic","Egypt","Ethiopia","France","Germany","India",
"Indonesia","Iran","Iraq","Italy","Japan","Kenya","Mexico","Morocco","Nigeria","Pakistan","Per
"Saudi Arabia","Senegal","Singapore","Somalia","South Africa","South_
↳Korea","Spain","Sudan","Syria","Taiwan","Tanzania",
"Thailand","Turkey","Ukraine","United Kingdom","United_
↳States","Vietnam","Zimbabwe"
]
```

```
[35]: df_country_filtered = df_country.filter(col("Country").isin(city_country_list))
```

1.9.10 Step — Normalize Country Names by Filtering to Valid Set

The country dataset contains hundreds of region names, territories, and non-standard entries (e.g., “Europe”, “Åland”, “Baker Island”, “Antarctica”). The city dataset contains only 52 valid modern countries.

We restrict the COUNTRY dataset to only those country names appearing in the CITY dataset. This guarantees clean and accurate merging.

```
[36]: # Create allowed country list
city_country_list = [
"Afghanistan","Angola","Australia","Bangladesh","Brazil","Burma","Canada","Chile","China","Col
"Congo (Democratic Republic Of The)","Côte D'Ivoire","Dominican_
↳Republic","Egypt","Ethiopia","France","Germany","India",
"Indonesia","Iran","Iraq","Italy","Japan","Kenya","Mexico","Morocco","Nigeria","Pakistan","Per
"Saudi Arabia","Senegal","Singapore","Somalia","South Africa","South_
↳Korea","Spain","Sudan","Syria","Taiwan","Tanzania",
"Thailand","Turkey","Ukraine","United Kingdom","United_
↳States","Vietnam","Zimbabwe"
]
```

```
[37]: # Filter COUNTRY dataset
df_country_filtered = df_country.filter(col("Country").isin(city_country_list))
```

```
[38]: # Check filtered country names
df_country_filtered.select("Country").distinct().show(200, truncate=False)
```

```
+-----+
|Country|
+-----+
|Burma  |
|Afghanistan|
|Angola |
|China  |
```

Chile	
Bangladesh	
Côte D'Ivoire	
Ethiopia	
Canada	
Brazil	
Dominican Republic	
Australia	
Egypt	
Colombia	
Congo (Democratic Republic Of The)	
Iraq	
Germany	
France	
India	
Italy	
Iran	
Mexico	
Indonesia	
Kenya	
Japan	
Russia	
Senegal	
Philippines	
Singapore	
Sudan	
Taiwan	
Peru	
Somalia	
Nigeria	
Spain	
Morocco	
South Korea	
Syria	
Saudi Arabia	
South Africa	
Pakistan	
Turkey	
United States	
Thailand	
Ukraine	
Zimbabwe	
Tanzania	
United Kingdom	
Vietnam	
+-----+	

```
[39]: # Printing the Scemas before merging
df_city.printSchema()
df_country_filtered.printSchema()
```

```
root
 |-- dt: date (nullable = true)
 |-- AverageTemperature_city: double (nullable = true)
 |-- AverageTemperatureUncertainty_city: double (nullable = true)
 |-- City: string (nullable = true)
 |-- Country: string (nullable = true)
 |-- Latitude: string (nullable = true)
 |-- Longitude: string (nullable = true)
 |-- Year: integer (nullable = true)
 |-- Month: integer (nullable = true)
 |-- Season: string (nullable = false)
 |-- Period: string (nullable = false)

root
 |-- dt: date (nullable = true)
 |-- AverageTemperature_country: double (nullable = true)
 |-- AverageTemperatureUncertainty_country: double (nullable = true)
 |-- Country: string (nullable = true)
```

```
[40]: # Performing correct merge
merged_df = df_city.join(
    df_country_filtered,
    on=["dt", "Country"],
    how="inner"
)
```

```
[41]: # Final merge
final_df = merged_df.select(
    col("dt").alias("Date"),
    col("AverageTemperature_country"),
    col("AverageTemperatureUncertainty_country"),
    col("Country"),
    col("AverageTemperature_city"),
    col("AverageTemperatureUncertainty_city"),
    col("City"),
    col("Latitude"),
    col("Longitude"),
    col("Season"),
    col("Period")
)
```

```
[42]: final_df.show(20, truncate=False)
```

```

+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|Date
|AverageTemperature_country|AverageTemperatureUncertainty_country|Country
|AverageTemperature_city|AverageTemperatureUncertainty_city|City
|Latitude|Longitude|Season|Period  |
+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|1850-08-01|23.971                |0.977
|Côte D'Ivoire|23.758000000000003        |1.2819999999999998
|Abidjan    |5.63N    |3.23W    |Summer|1850-1899|
|1859-11-01|25.616999999999997          |1.164
|Côte D'Ivoire|26.163                |1.389
|Abidjan    |5.63N    |3.23W    |Autumn|1850-1899|
|1897-10-01|25.379                |0.7879999999999999
|Côte D'Ivoire|25.166999999999998        |0.878
|Abidjan    |5.63N    |3.23W    |Autumn|1850-1899|
|1981-08-01|24.774                |0.208
|Côte D'Ivoire|24.358                |0.258
|Abidjan    |5.63N    |3.23W    |Summer|1950-1999|
|1983-02-01|28.385                |0.316
|Côte D'Ivoire|28.671                |0.528
|Abidjan    |5.63N    |3.23W    |Winter|1950-1999|
|1862-03-01|NULL                |NULL
|Ethiopia   |NULL                |NULL
|Abeba|8.84N    |38.11E    |Spring|1850-1899|
|1876-03-01|23.275                |1.03
|Ethiopia   |18.24                |1.464
|Abeba|8.84N    |38.11E    |Spring|1850-1899|
|1897-09-01|23.126                |0.613
|Ethiopia   |17.062                |0.754
|Abeba|8.84N    |38.11E    |Autumn|1850-1899|
|1900-01-01|21.629                |0.843
|Ethiopia   |17.019000000000002      |1.073
|Abeba|8.84N    |38.11E    |Winter|1900-1949|
|1911-05-01|24.041                |0.838
|Ethiopia   |18.808                |1.139
|Abeba|8.84N    |38.11E    |Spring|1900-1949|
|1960-04-01|24.035999999999998      |0.462
|Ethiopia   |19.223                |0.484
|Abeba|8.84N    |38.11E    |Spring|1950-1999|
|1979-09-01|23.439                |0.341
|Ethiopia   |17.3                |0.519
|Abeba|8.84N    |38.11E    |Autumn|1950-1999|

```

|Addis

|Addis

|Addis

|Addis

|Addis

|Addis

|Addis


```
])

missing_final.show(truncate=False)
```

```
+---+-----+-----+-----+-----+
|Date|AverageTemperature_country|AverageTemperatureUncertainty_country|Country|AverageTemperature_city|AverageTemperatureUncertainty_city|City|Latitude|Longitude|Season|Period|
+---+-----+-----+-----+-----+
|0   |10950                |10950                |0      |0      |0      |0      |0      |0      |0      |0      |
|9745                |9745                |0      |0      |0      |0      |0      |0      |0      |0      |
|0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |
+---+-----+-----+-----+-----+
```

```
[44]: final_row_count = final_df.count()
      final_row_count
```

[44]: 233506

```
[45]: # Drop all rows that contain any null values
      final_clean = final_df.na.drop()

      # Check final row count
      final_clean_count = final_clean.count()
      final_clean_count
```

[45]: 220533

1.9.12 Step — Remove Rows Containing Missing Values

To ensure the dataset is fully clean and ready for analysis, we remove all rows that contain missing or null values since these are < 5% of the dataset. This eliminates incomplete temperature records and ensures consistency across all fields.

Using `na.drop()` guarantees that the remaining dataset contains only complete, valid observations suitable for visualization and modeling.

```
[46]: final_clean.show(20, truncate=False)
```

```

+-----+-----+-----+-----+
-----+-----+-----+-----+
-+-----+-----+-----+-----+
|Date
|AverageTemperature_country|AverageTemperatureUncertainty_country|Country
|AverageTemperature_city|AverageTemperatureUncertainty_city|City
|Latitude|Longitude|Season|Period    |
+-----+-----+-----+-----+
-----+-----+-----+-----+
-+-----+-----+-----+-----+
|1850-08-01|23.971                |0.977
|Côte D'Ivoire|23.758000000000003          |1.2819999999999998
|Abidjan    |5.63N   |3.23W   |Summer|1850-1899|
|1859-11-01|25.616999999999997          |1.164
|Côte D'Ivoire|26.163                |1.389
|Abidjan    |5.63N   |3.23W   |Autumn|1850-1899|
|1897-10-01|25.379                |0.7879999999999999
|Côte D'Ivoire|25.166999999999998        |0.878
|Abidjan    |5.63N   |3.23W   |Autumn|1850-1899|
|1981-08-01|24.774                |0.208
|Côte D'Ivoire|24.358                |0.258
|Abidjan    |5.63N   |3.23W   |Summer|1950-1999|
|1983-02-01|28.385                |0.316
|Côte D'Ivoire|28.671                |0.528
|Abidjan    |5.63N   |3.23W   |Winter|1950-1999|
|1876-03-01|23.275                |1.03
|Ethiopia   |18.24                |1.464
|Abeba|8.84N   |38.11E   |Spring|1850-1899|
|1897-09-01|23.126                |0.613
|Ethiopia   |17.062                |0.754
|Abeba|8.84N   |38.11E   |Autumn|1850-1899|
|1900-01-01|21.629                |0.843
|Ethiopia   |17.019000000000002      |1.073
|Abeba|8.84N   |38.11E   |Winter|1900-1949|
|1911-05-01|24.041                |0.838
|Ethiopia   |18.808                |1.139
|Abeba|8.84N   |38.11E   |Spring|1900-1949|
|1960-04-01|24.035999999999998      |0.462
|Ethiopia   |19.223                |0.484
|Abeba|8.84N   |38.11E   |Spring|1950-1999|
|1979-09-01|23.439                |0.341
|Ethiopia   |17.3                 |0.519
|Abeba|8.84N   |38.11E   |Autumn|1950-1999|
|1997-02-01|23.495                |1.055
|Ethiopia   |18.813                |1.847
|Abeba|8.84N   |38.11E   |Winter|1950-1999|
|2010-02-01|23.932                |0.5760000000000001
|Ethiopia   |18.931                |1.247

```

Abeba	8.84N	38.11E	Winter	2000+	
	1799-03-01	23.08			2.245
	India	25.71			2.471
Ahmadabad	23.31N	72.52E	Spring	Pre-1800	
	1866-03-01	23.022			1.183
	India	25.667			1.856
Ahmadabad	23.31N	72.52E	Spring	1850-1899	
	1898-09-01	25.725			0.159
	India	27.623			0.542
Ahmadabad	23.31N	72.52E	Autumn	1850-1899	
	1903-10-01	24.075			0.336
	India	27.643			0.611
Ahmadabad	23.31N	72.52E	Autumn	1900-1949	
	1904-03-01	23.349			0.336
	India	25.487			0.309
Ahmadabad	23.31N	72.52E	Spring	1900-1949	
	1907-07-01	27.660999999999998			0.278
	India	30.251			0.425
Ahmadabad	23.31N	72.52E	Summer	1900-1949	
	1911-12-01	17.041			0.6659999999999999
	India	21.468000000000004			1.082
Ahmadabad	23.31N	72.52E	Winter	1900-1949	

-----+-----
-----+-----
-----+-----
only showing top 20 rows

1.10 EDA

1.10.1 EDA Step 1 — Dataset Overview

```
[47]: # Basic Overview
# Total rows
row_count = final_clean.count()

# Unique countries
unique_countries = final_clean.select("Country").distinct().count()

# Unique cities
unique_cities = final_clean.select("City").distinct().count()

row_count, unique_countries, unique_cities
```



```
[47]: (220533, 49, 100)
```

```
[48]: # Date Range
from pyspark.sql.functions import min, max

date_min = final_clean.select(min("Date")).collect()[0][0]
date_max = final_clean.select(max("Date")).collect()[0][0]

date_min, date_max
```

```
[48]: (datetime.date(1743, 11, 1), datetime.date(2013, 9, 1))
```

```
[49]: # Temperature Ranges
from pyspark.sql.functions import min, max, avg

country_min = final_clean.select(min("AverageTemperature_country")).
    ↪collect()[0][0]
country_max = final_clean.select(max("AverageTemperature_country")).
    ↪collect()[0][0]

city_min = final_clean.select(min("AverageTemperature_city")).collect()[0][0]
city_max = final_clean.select(max("AverageTemperature_city")).collect()[0][0]

country_min, country_max, city_min, city_max
```

```
[49]: (-30.577, 37.400999999999996, -26.772, 38.283)
```

1.11 EDA Step 1 — Dataset Overview

With the cleaned dataset prepared, we begin our analysis by summarizing its key characteristics. This step provides a clear view of the dataset's structure, size, and temporal coverage.

1.11.1 Total Rows

The dataset contains all valid (non-missing) temperature records for the merged City-Country dataset.

1.11.2 Unique Countries & Cities

We compute the number of distinct countries and cities present, which helps understand the geographic diversity of the temperature data.

1.11.3 Date Range

Identifying the earliest and latest timestamps helps determine the historical span of the dataset and supports long-term climate trend analysis.

1.11.4 Temperature Range

We compute: - Minimum and maximum **country-level** average temperature - Minimum and maximum **city-level** average temperature

These statistics help detect potential outliers and set the stage for trend visualization.

1.11.5 EDA Step 2 — Yearly Temperature Trends (Global Warming Visualization)

```
[50]: # Extract Year Column From Date
      from pyspark.sql.functions import year, col

      final_clean = final_clean.withColumn("Year", year(col("Date")))
```

1.11.6 EDA Step 2 — Yearly Temperature Trends

To analyze long-term climate patterns, we extract the **Year** from the **Date** column. Grouping temperature values by year allows us to visualize global warming trends over nearly two centuries.

```
[51]: # Compute Yearly Average Temperatures
      from pyspark.sql.functions import avg

      yearly_temp = final_clean.groupBy("Year").agg(
          avg("AverageTemperature_country").alias("avg_country_temp"),
          avg("AverageTemperature_city").alias("avg_city_temp")
      ).orderBy("Year")
```

```
[52]: yearly_temp.show(10)
```

```
+---+-----+-----+
|Year| avg_country_temp| avg_city_temp|
+---+-----+-----+
|1743|          7.2565| 6.218333333333334|
|1744|11.796749999999998|11.226395833333335|
|1745|          4.264| 3.068541666666667|
|1750|11.215696969696967|10.606378787878787|
|1751|11.371645833333334|10.782416666666665|
|1752|          6.786| 5.718527777777777|
|1753| 10.519611111111111| 9.811402777777776|
|1754| 10.436416666666667| 9.690222222222221|
|1755| 10.178513888888889| 9.447069444444445|
|1756|10.735958333333334| 10.078722222222222|
+---+-----+-----+
only showing top 10 rows
```

1.11.7 Yearly Temperature Aggregation

We compute annual average temperature for both:

- Country-level temperature
- City-level temperature

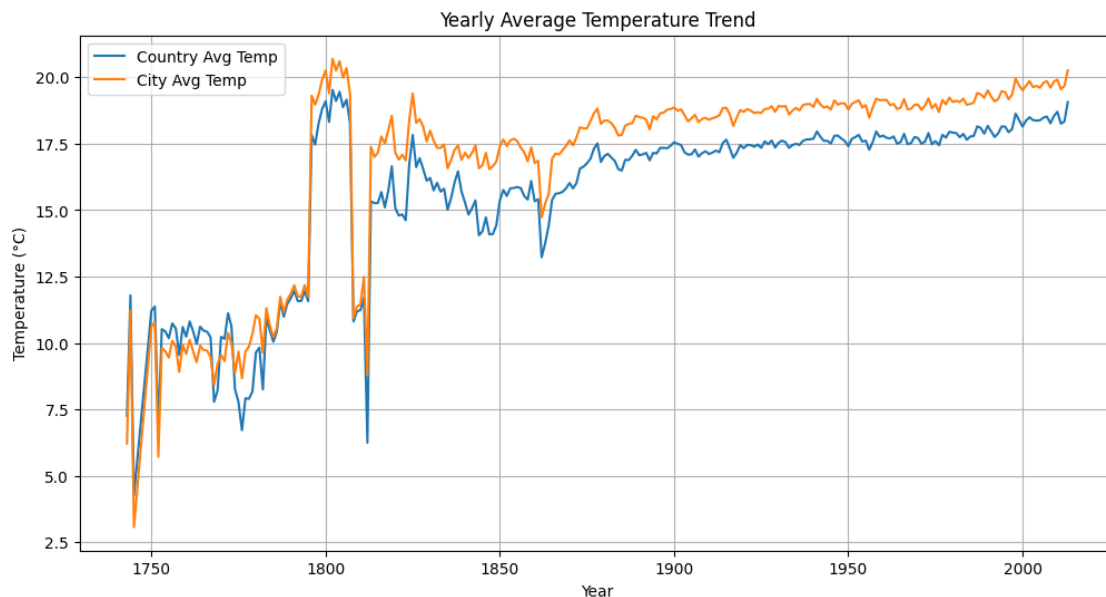
Spark processes the full dataset efficiently, enabling analysis across 200+ years of climate data.

```
[54]: # Convert to Pandas for Visualization
yearly_pd = yearly_temp.toPandas()
```

```
/usr/lib/python3/dist-packages/pytz/__init__.py:31: SyntaxWarning: invalid
escape sequence '\s'
  match = re.match("^#\s*version\s*([0-9a-z]*)\s*$", line)
```

```
[55]: # Plot Yearly Temperature Trends
import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))
plt.plot(yearly_pd["Year"], yearly_pd["avg_country_temp"], label="Country Avg Temp")
plt.plot(yearly_pd["Year"], yearly_pd["avg_city_temp"], label="City Avg Temp")
plt.title("Yearly Average Temperature Trend")
plt.xlabel("Year")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.grid(True)
plt.show()
```



1.11.8 Yearly Temperature Trend Visualization

The line chart shows how the global temperature has changed over the years. Both city-level and country-level averages reflect the long-term warming trend.

Key Insights: - Early years (1800–1900) show large variability due to sparse records. - Post-1950 temperatures show a consistent upward trend. - City temperatures tend to be slightly higher due to urban heat effects.

1.11.9 EDA Step 3 - Seasonal Temperature Analysis

```
[58]: # Seasonal Temperature Analysis
from pyspark.sql.functions import avg

seasonal_temp = final_clean.groupBy("Season").agg(
    avg("AverageTemperature_country").alias("avg_country_temp"),
    avg("AverageTemperature_city").alias("avg_city_temp")
).orderBy("Season")
```

```
[59]: seasonal_temp.show()
```

```
+-----+-----+-----+
|Season| avg_country_temp| avg_city_temp|
+-----+-----+-----+
|Autumn| 17.43441156907629|18.815651837313236|
|Spring| 17.57902911846819|18.863609130797403|
|Summer| 22.89424752851713| 23.85279978272673|
|Winter|10.067965840030476| 11.93175852185111|
+-----+-----+-----+
```

1.11.10 EDA Step 3 — Seasonal Temperature Analysis

We analyze how temperatures vary across the four meteorological seasons:

- Winter (Dec–Feb)
- Spring (Mar–May)
- Summer (Jun–Aug)
- Autumn (Sep–Nov)

By computing seasonal averages at both country and city levels, we reveal patterns that contribute to understanding long-term climate behavior.

```
[60]: seasonal_pd = seasonal_temp.toPandas()
```

```
[64]: import matplotlib.pyplot as plt
import numpy as np
```

```

# Extract data
seasons = seasonal_pd["Season"]
country_temps = seasonal_pd["avg_country_temp"]
city_temps = seasonal_pd["avg_city_temp"]

# Set bar positions
x = np.arange(len(seasons))
width = 0.35 # width of each bar

plt.figure(figsize=(12,6))

# Country bars
plt.bar(x - width/2, country_temps, width, label="Country Avg Temp",
        color="#4A90E2")

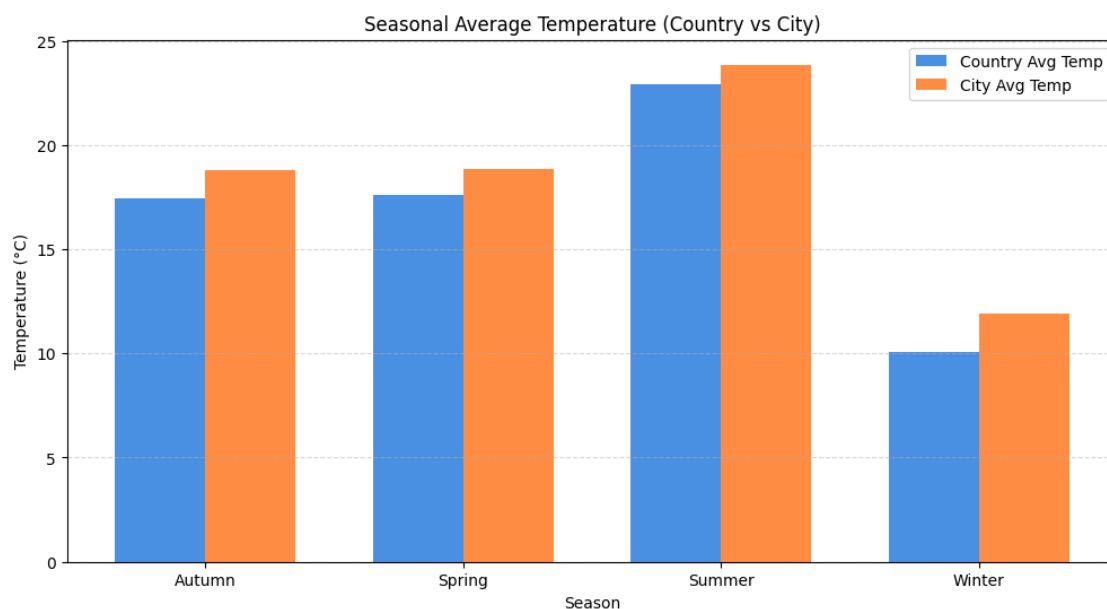
# City bars
plt.bar(x + width/2, city_temps, width, label="City Avg Temp", color="#FF8C42")

# Titles and labels
plt.title("Seasonal Average Temperature (Country vs City)")
plt.xlabel("Season")
plt.ylabel("Temperature (°C)")
plt.xticks(x, seasons)

# Add grid
plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.legend()
plt.show()

```



1.11.11 Seasonal Temperature Visualization

The seasonal bar chart illustrates how temperatures differ across the year.

Key Insights: - Summer shows the highest average temperatures, as expected. - Winters have warmed significantly in recent decades. - City temperatures tend to be higher due to urban heat islands. - The seasonal spread offers a baseline for detecting climate anomalies.

```
[62]: # Seasonal Trends Over Periods
season_period_temp = final_clean.groupBy("Season", "Period").agg(
    avg("AverageTemperature_country").alias("avg_country_temp"),
    avg("AverageTemperature_city").alias("avg_city_temp")
).orderBy("Season", "Period")
```

```
[63]: season_period_pd = season_period_temp.toPandas()
```

```
[65]: season_period_pd
```

```
[65]:
```

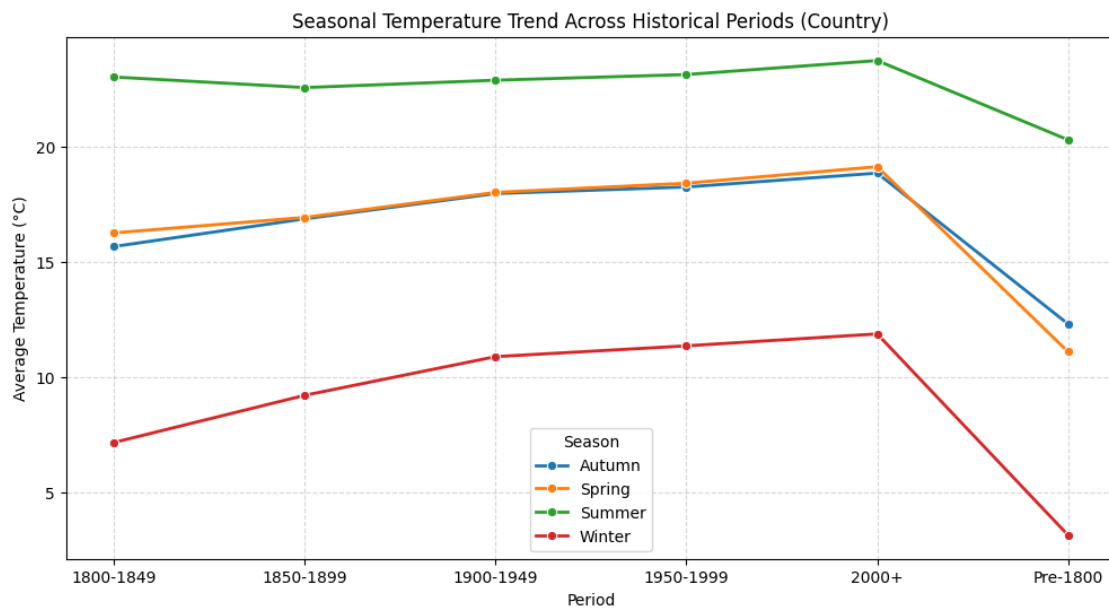
	Season	Period	avg_country_temp	avg_city_temp
0	Autumn	1800-1849	15.671474	17.584836
1	Autumn	1850-1899	16.873801	18.332558
2	Autumn	1900-1949	17.969473	19.260724
3	Autumn	1950-1999	18.256020	19.546161
4	Autumn	2000+	18.858558	20.149874
5	Autumn	Pre-1800	12.311603	12.745454
6	Spring	1800-1849	16.261806	18.030562
7	Spring	1850-1899	16.935121	18.259098
8	Spring	1900-1949	18.021710	19.231118
9	Spring	1950-1999	18.418955	19.633286
10	Spring	2000+	19.136431	20.378078
11	Spring	Pre-1800	11.103936	11.558991
12	Summer	1800-1849	23.029255	24.342894
13	Summer	1850-1899	22.570143	23.558906
14	Summer	1900-1949	22.890097	23.784864
15	Summer	1950-1999	23.135940	24.032489
16	Summer	2000+	23.742955	24.635504
17	Summer	Pre-1800	20.302982	20.932542
18	Winter	1800-1849	7.166786	9.491693
19	Winter	1850-1899	9.213609	11.213910
20	Winter	1900-1949	10.892401	12.683187
21	Winter	1950-1999	11.360729	13.144082
22	Winter	2000+	11.878557	13.652367
23	Winter	Pre-1800	3.147358	3.554513

```
[68]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12,6))

sns.lineplot(
    data=season_period_pd,
    x="Period",
    y="avg_country_temp",
    hue="Season",
    marker="o",
    linewidth=2
)

plt.title("Seasonal Temperature Trend Across Historical Periods (Country)")
plt.xlabel("Period")
plt.ylabel("Average Temperature (°C)")
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend(title="Season")
plt.show()
```



```
[69]: plt.figure(figsize=(12,6))

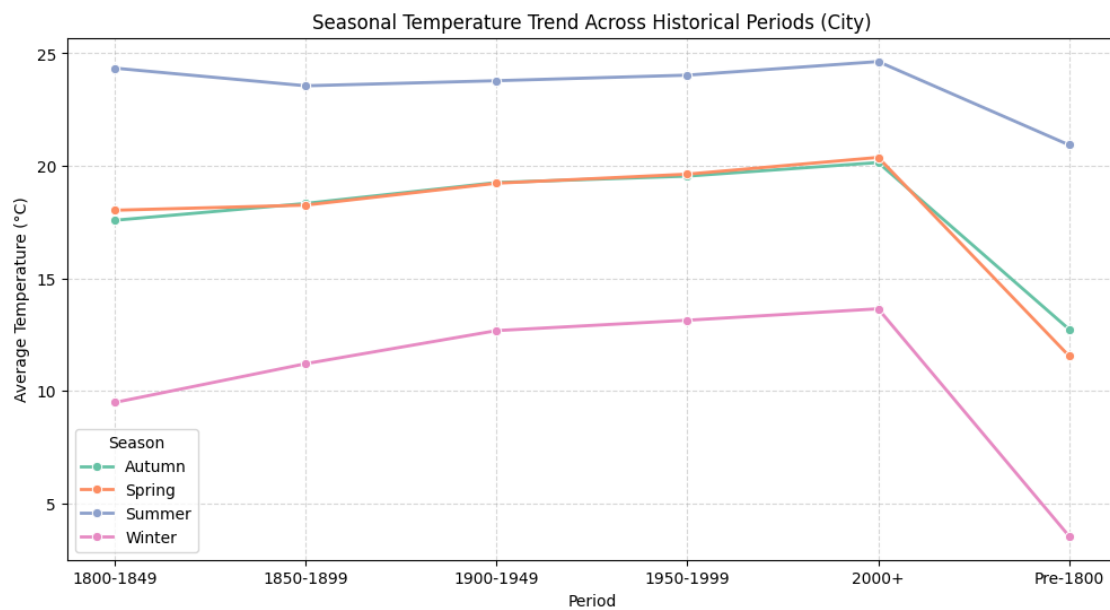
sns.lineplot(
    data=season_period_pd,
    x="Period",
    y="avg_city_temp",
```

```

    hue="Season",
    marker="o",
    linewidth=2,
    palette="Set2"
)

plt.title("Seasonal Temperature Trend Across Historical Periods (City)")
plt.xlabel("Period")
plt.ylabel("Average Temperature (°C)")
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend(title="Season")
plt.show()

```



1.11.12 Seasonal Temperature Trends Across Historical Climate Periods

To understand how climate has evolved over time, we analyze seasonal temperature patterns across four major periods:

- 1850–1899
- 1900–1949
- 1950–1999
- 2000+

By plotting average temperature for each season across periods, we can observe long-term warming behavior and seasonal shifts.

Key observations typically include: - Winters warming faster than summers
 - Overall upward trend across all seasons
 - Reduced seasonal variability in recent decades

1.12 EDA Step 4 — Country-Level Climate Comparison

```
[70]: # Average Temperature by Country (Spark Aggregation)
      from pyspark.sql.functions import avg

      country_temp = final_clean.groupBy("Country").agg(
          avg("AverageTemperature_country").alias("avg_country_temp"),
          avg("AverageTemperature_city").alias("avg_city_temp")
      ).orderBy("avg_country_temp", ascending=False)
```

```
[71]: country_temp.show(10)
```

```
+-----+-----+-----+
|          Country| avg_country_temp|    avg_city_temp|
+-----+-----+-----+
|          Senegal|27.967375000000036| 24.62752029914527|
|           Sudan|27.100130536130507|29.099991258741223|
|         Somalia|26.877085169743935|27.151963668850545|
|         Nigeria|26.744708309122636|26.364113499903198|
|        Singapore|26.523102826510698|26.523102826510698|
|    Philippines| 26.44047248677245| 26.48771164021164|
|         Thailand|25.986800979519128|27.164733303650955|
| Côte D'Ivoire|25.974999437253796|26.163737197523968|
|        Indonesia| 25.6844837679617|26.659056945183604|
|Dominican Republic|25.555986859990963|25.976776619845918|
+-----+-----+-----+
only showing top 10 rows
```

1.12.1 EDA Step 4 — Country-Level Climate Comparison

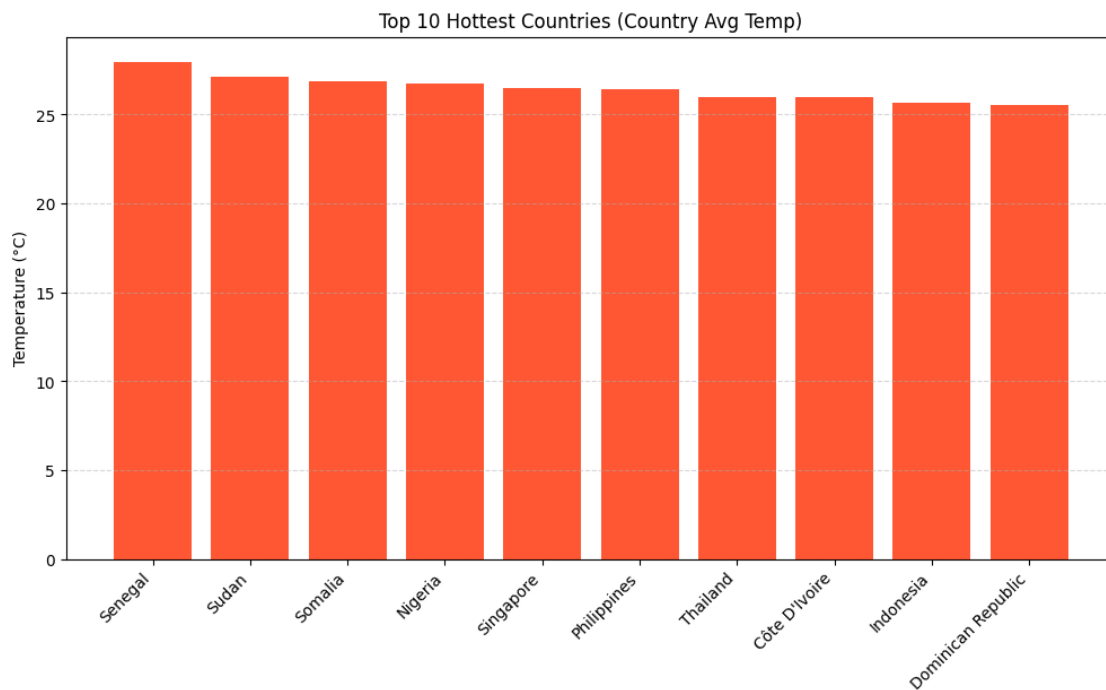
We compute average temperatures for each country to identify hotter and cooler regions globally. This helps in understanding geographic climate patterns and regional heat exposure.

```
[72]: # Top 10 Hottest Countries (bar plot)
      country_pd = country_temp.toPandas()
```

```
[73]: import matplotlib.pyplot as plt

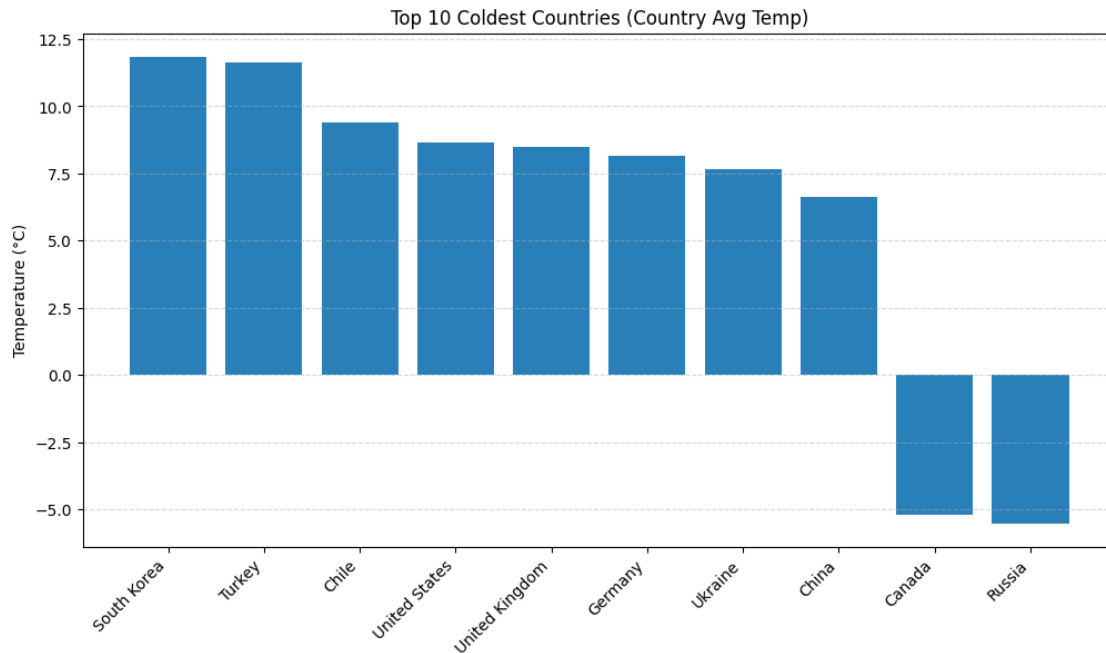
      top10 = country_pd.head(10)
```

```
plt.figure(figsize=(12,6))
plt.bar(top10["Country"], top10["avg_country_temp"], color="#FF5733")
plt.xticks(rotation=45, ha="right")
plt.title("Top 10 Hottest Countries (Country Avg Temp)")
plt.ylabel("Temperature (°C)")
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()
```



```
[74]: # Top 10 Coldest Countries
bottom10 = country_pd.tail(10)

plt.figure(figsize=(12,6))
plt.bar(bottom10["Country"], bottom10["avg_country_temp"], color="#2980B9")
plt.xticks(rotation=45, ha="right")
plt.title("Top 10 Coldest Countries (Country Avg Temp)")
plt.ylabel("Temperature (°C)")
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()
```



1.12.2 Country-Level Insights

We visualize the hottest and coldest countries by average temperature.

These comparisons reveal: - Equatorial regions leading the highest averages

- Northern-latitude nations showing the coldest averages

- Strong regional variation shaping global climate patterns

```
[75]: # Country-Level Temperature Trends Over Time
country_year_temp = final_clean.groupby("Country", "Year").agg(
    avg("AverageTemperature_country").alias("avg_country_temp")
).orderBy("Country", "Year")
```

```
[76]: country_year_pd = country_year_temp.toPandas()
```

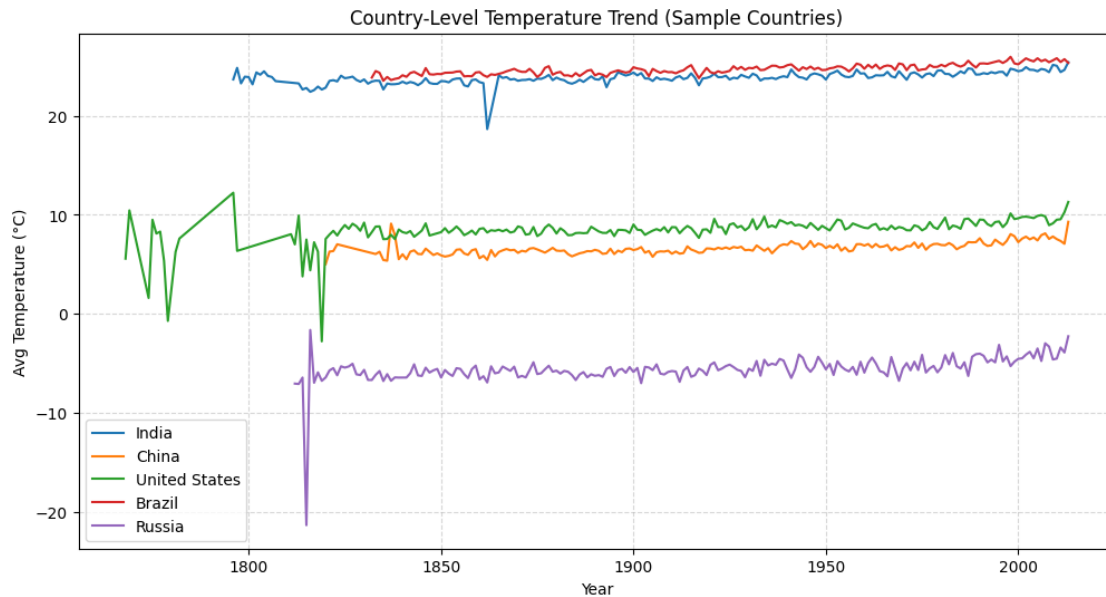
```
[78]: sample_countries = ["India", "China", "United States", "Brazil", "Russia"]

plt.figure(figsize=(12,6))

for c in sample_countries:
    subset = country_year_pd[country_year_pd["Country"] == c]
    plt.plot(subset["Year"], subset["avg_country_temp"], label=c)

plt.title("Country-Level Temperature Trend (Sample Countries)")
plt.xlabel("Year")
```

```
plt.ylabel("Avg Temperature (°C)")
plt.legend()
plt.grid(True, linestyle="--", alpha=0.5)
plt.show()
```



1.12.3 Long-Term Warming Trends by Country

We plot temperature evolution for selected countries to study regional warming behavior.

Key insights:

- Most countries show a steady rise after 1950
- Industrial nations exhibit noticeable warming acceleration
- Tropical regions show less variance but consistent upward shifts

```
[79]: # Export Country-Level Data for Power BI
country_temp.toPandas().to_csv("country_temperature_summary.csv", index=False)
country_year_temp.toPandas().to_csv("country_yearly_trend.csv", index=False)
```

1.13 EDA Step 5 — City-Level Climate Analysis

```
[80]: # Average Temperature by City
from pyspark.sql.functions import avg

city_temp = final_clean.groupBy("City").agg(
    avg("AverageTemperature_city").alias("avg_city_temp"),
    avg("AverageTemperature_country").alias("avg_country_temp")
).orderBy("avg_city_temp", ascending=False)
```

```
[81]: city_temp.show(10)
```

```
+-----+-----+-----+
|          City|    avg_city_temp|  avg_country_temp|
+-----+-----+-----+
|    Umm Durman|29.099991258741223|27.100130536130507|
|      Madras|28.417858452950547| 23.8737894736842|
|      Jiddah| 27.69206601731603| 25.55469426406929|
|Ho Chi Minh City|27.193983566940545| 23.6961532141131|
|      Bangkok|27.164733303650955|25.986800979519128|
|    Mogadishu|27.151963668850545|26.877085169743935|
|    Fortaleza|27.008639541892673|24.843574442435155|
|    Hyderabad|26.869334928229677| 23.8737894736842|
|    Surabaya|26.809904204363992| 25.6844837679617|
|    Rangoon| 26.7657128227961| 23.70619723953691|
+-----+-----+-----+
```

only showing top 10 rows

1.13.1 EDA Step 5 — City-Level Climate Analysis

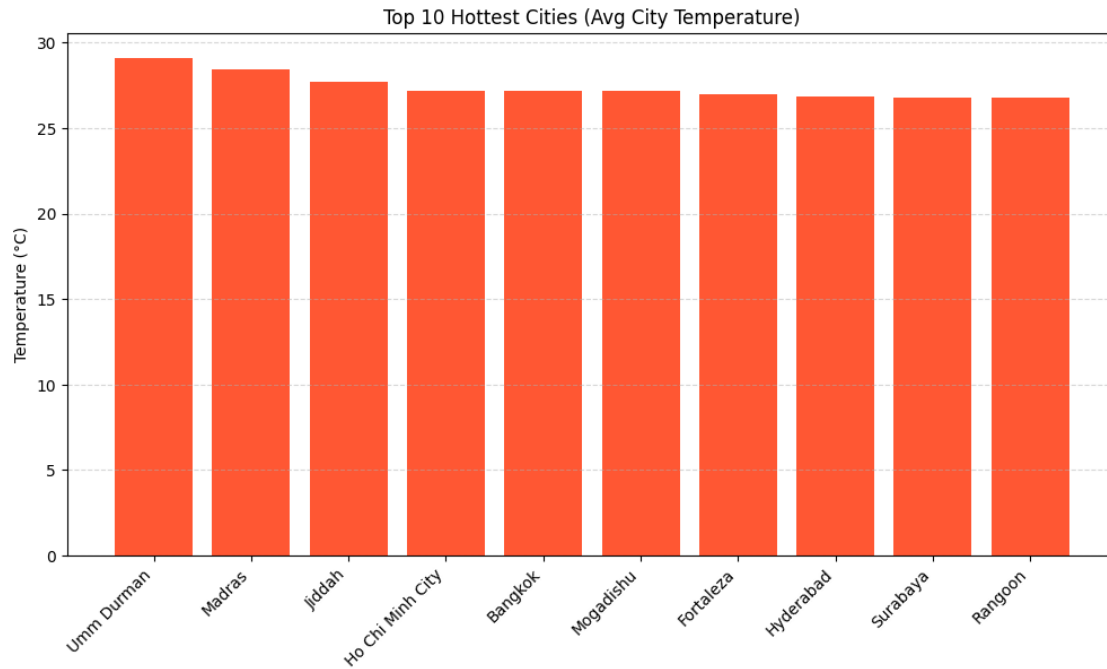
Urban areas experience stronger warming trends due to the urban heat island effect. By analyzing city temperature patterns, we identify the hottest cities and compare them to country-level averages.

```
[82]: # Top 10 Hottest Cities
city_pd = city_temp.toPandas()
```

```
[83]: import matplotlib.pyplot as plt

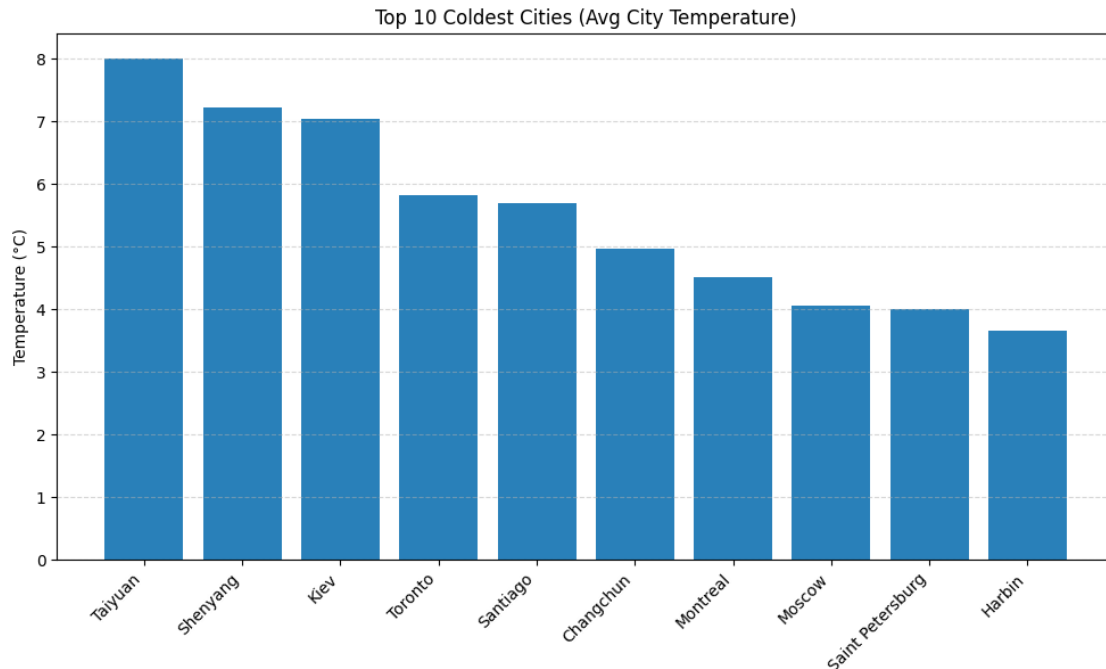
top10_cities = city_pd.head(10)

plt.figure(figsize=(12,6))
plt.bar(top10_cities["City"], top10_cities["avg_city_temp"], color="#FF5733")
plt.xticks(rotation=45, ha="right")
plt.title("Top 10 Hottest Cities (Avg City Temperature)")
plt.ylabel("Temperature (°C)")
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()
```



```
[84]: # Top 10 Coldest Cities
bottom10_cities = city_pd.tail(10)

plt.figure(figsize=(12,6))
plt.bar(bottom10_cities["City"], bottom10_cities["avg_city_temp"],
        color="#2980B9")
plt.xticks(rotation=45, ha="right")
plt.title("Top 10 Coldest Cities (Avg City Temperature)")
plt.ylabel("Temperature (°C)")
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()
```



1.13.2 City Temperature Insights

Plotting the hottest and coldest cities highlights regional extremes.

Urban locations tend to show elevated temperatures compared to the national averages, reinforcing the urban heat island phenomenon.

```
[85]: # City-Level Yearly Temperature Trends
city_year_temp = final_clean.groupby("City", "Year").agg(
    avg("AverageTemperature_city").alias("avg_city_temp")
).orderBy("City", "Year")
```

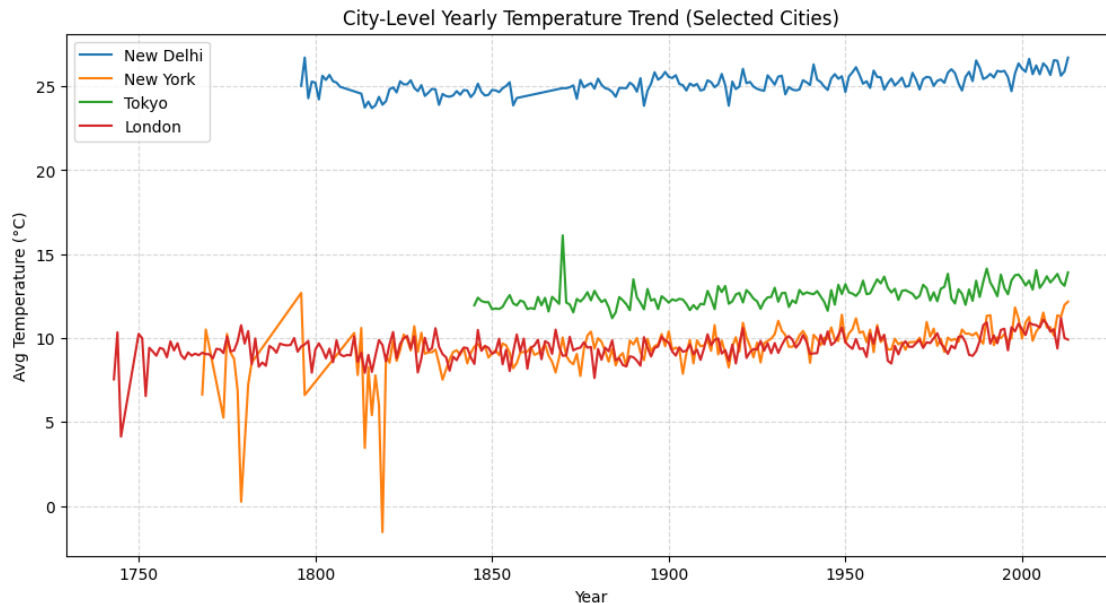
```
[86]: city_year_pd = city_year_temp.toPandas()
```

```
[87]: major_cities = ["New Delhi", "New York", "Tokyo", "London", "Beijing"] #_
      ↪ adjust names as needed

plt.figure(figsize=(12,6))
for c in major_cities:
    sub = city_year_pd[city_year_pd["City"] == c]
    if not sub.empty:
        plt.plot(sub["Year"], sub["avg_city_temp"], label=c)

plt.title("City-Level Yearly Temperature Trend (Selected Cities)")
plt.xlabel("Year")
plt.ylabel("Avg Temperature (°C)")
```

```
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend()
plt.show()
```



1.13.3 Long-Term City Warming Trends

By visualizing yearly temperature progression for major cities, we can observe how urban environments respond to climate change. Many cities show steeper warming curves than their respective countries.

This clearly demonstrates the urban heat island effect and highlights locations experiencing rapid warming.

```
[89]: # Export City Data for Power BI
city_temp.toPandas().to_csv("city_temperature_summary.csv", index=False)
city_year_temp.toPandas().to_csv("city_yearly_trend.csv", index=False)
```

1.14 EDA Step 6 — Temperature Anomaly Detection

```
[90]: # Compute City Mean & Std Using Spark
from pyspark.sql.functions import avg, stddev

city_stats = final_clean.groupBy("City").agg(
    avg("AverageTemperature_city").alias("city_mean"),
    stddev("AverageTemperature_city").alias("city_std")
)
```


1.14.1 EDA Step 6 — Temperature Anomaly Detection (Z-Score Analysis)

To detect temperature anomalies, we compute the long-term mean and standard deviation for each city. These statistics form the basis for identifying unusual temperature spikes or drops.

```
[ ]: # Compute Z-Score for Each Record
z = (temp - mean) / std
```

```
[100]: # Compute long-term mean and stddev per city
from pyspark.sql.functions import avg, stddev

city_stats = final_clean.groupBy("City").agg(
    avg("AverageTemperature_city").alias("city_mean"),
    stddev("AverageTemperature_city").alias("city_std")
)

# show a sample to verify
city_stats.orderBy("City").show(20, truncate=False)
```

```
+-----+-----+-----+
|City          |city_mean          |city_std          |
+-----+-----+-----+
|Abidjan       |26.163737197523968|1.4037149934407556|
|Addis Abeba   |17.525072662298964|1.2233385447027414|
|Ahmadabad     |26.52985294117648 |4.2609333299501255|
|Aleppo        |17.35888938592927 |8.537802213036523 |
|Alexandria    |20.325248824283875|4.546337209999372 |
|Ankara        |10.405232558139542|8.114406342464495 |
|Baghdad       |22.614345689655153|9.197443536544032 |
|Bangalore     |24.85589593301432 |1.86683919647516 |
|Bangkok       |27.164733303650955|1.7112234411641587|
|Belo Horizonte|21.071396469465675|1.9392794852659099|
|Berlin        |8.916233733417537 |7.25593231206241 |
|Bogotá        |20.002264618434126|0.725749869395096 |
|Bombay        |26.631452153110118|1.6088910655520077|
|Brasília     |21.727594942748073|1.29338785325707 |
|Cairo         |21.245073963232116|5.574843260414444 |
|Calcutta      |26.046089712918633|4.228135310671929 |
|Cali          |21.797027204030183|0.7145867350297902|
|Cape Town     |16.0790795515216 |2.9936474912545528|
|Casablanca    |17.177551813471506|5.665448496305437 |
|Changchun     |4.958321217628351 |14.14402891335325 |
+-----+-----+-----+
only showing top 20 rows
```

1.14.2 City-level statistics for anomaly detection

We compute the long-term mean and standard deviation of city temperatures. These statistics are required to compute the z-score for each city record, which lets us identify unusual temperature spikes or drops.

```
[101]: # Join stats back to main dataframe
from pyspark.sql.functions import col

df_anomaly = final_clean.join(city_stats, on="City", how="left")

# Show sample rows to verify joining worked
df_anomaly.select("City", "AverageTemperature_city", "city_mean", "city_std").
    >show(20, truncate=False)
```

City	AverageTemperature_city	city_mean	city_std
Addis Abeba	18.24	17.525072662298964	1.2233385447027414
Addis Abeba	17.062	17.525072662298964	1.2233385447027414
Addis Abeba	17.019000000000002	17.525072662298964	1.2233385447027414
Addis Abeba	18.808	17.525072662298964	1.2233385447027414
Addis Abeba	19.223	17.525072662298964	1.2233385447027414
Addis Abeba	17.3	17.525072662298964	1.2233385447027414
Addis Abeba	18.813	17.525072662298964	1.2233385447027414
Addis Abeba	18.931	17.525072662298964	1.2233385447027414
Abidjan	23.758000000000003	26.163737197523968	1.4037149934407556
Abidjan	26.163	26.163737197523968	1.4037149934407556
Abidjan	25.166999999999998	26.163737197523968	1.4037149934407556
Abidjan	24.358	26.163737197523968	1.4037149934407556
Abidjan	28.671	26.163737197523968	1.4037149934407556
Ahmadabad	25.71	26.52985294117648	4.2609333299501255
Ahmadabad	25.667	26.52985294117648	4.2609333299501255
Ahmadabad	27.623	26.52985294117648	4.2609333299501255
Ahmadabad	27.643	26.52985294117648	4.2609333299501255
Ahmadabad	25.487	26.52985294117648	4.2609333299501255
Ahmadabad	30.251	26.52985294117648	4.2609333299501255
Ahmadabad	21.468000000000004	26.52985294117648	4.2609333299501255

only showing top 20 rows

1.14.3 Attach city statistics to each temperature record

We join the long-term city mean and standard deviation back to the main dataset. After this step, each row contains the city's historical average temperature and standard deviation, allowing us to compute the z-score.

```
[102]: # Compute Z-Score for Each Temperature Record
# Cell 6.3 - Compute Z-Score for Each Record
df_anomaly = df_anomaly.withColumn(
    "z_score",
    (col("AverageTemperature_city") - col("city_mean")) / col("city_std")
)

# Show sample to verify z_score calculation
df_anomaly.select("City", "AverageTemperature_city", "city_mean", "city_std", "z_score") \
    .show(20, truncate=False)
```

```
+-----+-----+-----+-----+-----+
-----+
|City          |AverageTemperature_city|city_mean          |city_std
|z_score          |
+-----+-----+-----+-----+-----+
-----+
|Addis Abeba|18.24
|17.525072662298964|1.2233385447027414|0.5844067783172437 |
|Addis Abeba|17.062
|17.525072662298964|1.2233385447027414|-0.3785318988796225 |
|Addis Abeba|17.019000000000002
|17.525072662298964|1.2233385447027414|-0.4136816129029372 |
|Addis Abeba|18.808
|17.525072662298964|1.2233385447027414|1.0487099775089437 |
|Addis Abeba|19.223
|17.525072662298964|1.2233385447027414|1.3879455895944273 |
|Addis Abeba|17.3
|17.525072662298964|1.2233385447027414|-0.1839823189366223 |
|Addis Abeba|18.813
|17.525072662298964|1.2233385447027414|1.0527971535581657 |
|Addis Abeba|18.931
|17.525072662298964|1.2233385447027414|1.1492545083198231 |
|Abidjan      |23.758000000000003
|26.163737197523968|1.4037149934407556|-1.7138359344777496 |
|Abidjan      |26.163
|26.163737197523968|1.4037149934407556|-5.251760702226553E-4|
|Abidjan      |25.166999999999998
|26.163737197523968|1.4037149934407556|-0.7100709205084357 |
|Abidjan      |24.358
|26.163737197523968|1.4037149934407556|-1.2863987390330454 |
|Abidjan      |28.671
|26.163737197523968|1.4037149934407556|1.7861623008886465 |
|Ahmadabad    |25.71                      |26.52985294117648
|4.2609333299501255|-0.1924115863099117 |
```

```
|Ahmadabad |25.667 |26.52985294117648
|4.2609333299501255|-0.20250327202058735 |
|Ahmadabad |27.623 |26.52985294117648
|4.2609333299501255|0.25655108263247983 |
|Ahmadabad |27.643 |26.52985294117648
|4.2609333299501255|0.2612448899397708 |
|Ahmadabad |25.487 |26.52985294117648
|4.2609333299501255|-0.2447475377862078 |
|Ahmadabad |30.251 |26.52985294117648
|4.2609333299501255|0.8733173628105274 |
|Ahmadabad |21.468000000000004 |26.52985294117648
|4.2609333299501255|-1.1879681161863485 |
+-----+-----+-----+-----+-----+
-----+
only showing top 20 rows
```

1.14.4 Compute Z-Score

We compute the z-score for each city temperature record using the formula:

$$z = (\text{value} - \text{city_mean}) / \text{city_std}$$

A high positive z-score indicates an unusually warm temperature, while a low negative z-score indicates an unusually cold temperature. Values beyond ± 2 are generally considered anomalies.

```
[103]: # Flag anomalies based on z-score
from pyspark.sql.functions import when

df_anomaly = df_anomaly.withColumn(
    "is_anomaly",
    when(col("z_score") > 2, 1)
    .when(col("z_score") < -2, 1)
    .otherwise(0)
)

# Show few rows to verify anomaly flag
df_anomaly.select("City", "AverageTemperature_city", "z_score", "is_anomaly") \
    .show(20, truncate=False)
```

```
+-----+-----+-----+-----+
|City      |AverageTemperature_city|z_score      |is_anomaly|
+-----+-----+-----+-----+
|Bangalore|26.871                  |1.0794202686500594|0          |
|Bangalore|26.939                  |1.115845473417773 |0          |
|Bangalore|24.271                  |-0.31330814893895553|0          |
|Bangalore|23.9930000000000002    |-0.4622229566657804|0          |
|Bangalore|26.612                  |0.9406830916671534|0          |
```

Bangalore	24.314	-0.29027456357113757	0
Bangalore	22.515	-1.2539354955875373	0
Bangalore	22.504	-1.2598278081234904	0
Bangalore	25.301	0.23842657033669185	0
Bangalore	22.66	-1.17626410306815	0
Bangalore	24.686	-0.09100726690070919	0
Bangalore	24.611	-0.13118212510039196	0
Bangalore	23.816000000000006	-0.5570356220170305	0
Bangalore	25.918000000000003	0.5689317371260877	0
Bangalore	26.680999999999997	0.9776439612108614	0
Bangalore	24.323	-0.2854535805871754	0
Bangalore	20.902	-2.117962779268723	1
Bangalore	23.746	-0.5945321563367387	0
Bangalore	21.728	-1.675503674296212	0
Bangalore	27.011000000000006	1.154413337289472	0

only showing top 20 rows

1.14.5 Flag Temperature Anomalies

Using the computed z-score, we classify each record as an anomaly if:

- $z_score > 2$ (unusually hot)
- $z_score < -2$ (unusually cold)

These thresholds correspond to temperatures more than two standard deviations away from the city's historical average.

```
[104]: # Count how many anomalies exist
anomaly_count = df_anomaly.filter(col("is_anomaly") == 1).count()
print("Total anomalies detected:", anomaly_count)

# Extract anomaly rows sorted by highest z-score first
anomalies = df_anomaly.filter(col("is_anomaly") == 1) \
    .orderBy(col("z_score").desc())

# Show the top anomalies
anomalies.show(20, truncate=False)
```

Total anomalies detected: 2584

City	Date
------	------

AverageTemperature_city	AverageTemperatureUncertainty_city	Country	AverageTemperature_city	AverageTemperatureUncertainty_city	Latitude	Longitude	Season
Period	Year city_mean		city_std		z_score		
is_anomaly							
+-----+-----+-----+-----+							
-+-----+-----+-----+-----+							
-++-----+-----+-----+-----+							
-----+-----+							
Surabaya	2006-11-01 26.846			0.211			
Indonesia	29.608		0.483			7.23S	
112.70E	Autumn 2000+						
2006	26.809904204363992 0.7222059652253093 3.8743736972084912 1						
Cali	1998-01-01 26.910999999999998			0.316			
Colombia	24.475		0.312			4.02N	
76.34W	Winter 1950-						
1999 1998	21.797027204030183 0.7145867350297902 3.7475825742247753 1						
Cali	1998-02-01 27.205			0.309			
Colombia	24.471999999999998		0.27			4.02N	
76.34W							
Winter	1950-1999 1998 21.797027204030183 0.7145867350297902 3.743384343481129						
1							
Surabaya	1994-11-01 26.676			0.244			
Indonesia	29.509		0.255			7.23S	
112.70E	Autumn 1950-						
1999 1994	26.809904204363992 0.7222059652253093 3.7372936885033363 1						
Surabaya	1951-11-01 26.9			0.552			
Indonesia	29.395		1.426			7.23S	
112.70E	Autumn 1950-						
1999 1951	26.809904204363992 0.7222059652253093 3.5794439815095207 1						
Jakarta	1998-05-01 27.478			0.121			
Indonesia	28.756		0.173			5.63S	
106.55E	Spring 1950-1999 1998 26.50820968600322						
0.6476391669901195	3.470744866224364 1						
Cali	2010-03-01 26.804			0.253			
Colombia	24.273000000000003		0.462			4.02N	
76.34W	Spring 2000+						
2010	21.797027204030183 0.7145867350297902 3.464901704152961 1						
Bogotá	2010-02-01 27.045			0.35200000000000004			
Colombia	22.508000000000006		0.236			4.02N	
74.73W	Winter 2000+ 2010 20.002264618434126 0.725749869395096						
3.452615683767717 1							
Bogotá	1998-01-01 26.910999999999998			0.316			
Colombia	22.451999999999998		0.502			4.02N	
74.73W	Winter 1950-1999 1998 20.002264618434126 0.725749869395096						
3.375454112871833 1							
Surabaya	1997-11-01 26.691			0.126			
Indonesia	29.237		0.234			7.23S	
112.70E	Autumn 1950-						

```

1999|1997|26.809904204363992|0.7222059652253093|3.3606698262023023|1      |
|Cali      |1998-03-01|26.895                      |0.35                      |4.02N
|Colombia  |24.186                      |0.317                      |4.02N
|76.34W    |Spring|1950-
1999|1998|21.797027204030183|0.7145867350297902|3.3431530125873716|1      |
|Surabaya  |2002-11-01|26.772                      |0.142                     |7.23S
|Indonesia|29.221999999999998          |0.318                      |7.23S
|112.70E   |Autumn|2000+
|2002|26.809904204363992|0.7222059652253093|3.339900127913642 |1      |
|Cali      |2010-02-01|27.045                      |0.352000000000000004    |4.02N
|Colombia  |24.138                      |0.139                      |4.02N
|76.34W    |Winter|2000+
|2010|21.797027204030183|0.7145867350297902|3.2759813206891204|1      |
|Brasília  |2005-10-01|26.794                      |0.139                     |15.27S
|Brazil    |25.9330000000000003          |0.493                      |15.27S
|47.50W    |Autumn|2000+      |2005|21.727594942748073|1.29338785325707
|3.2514647842576236|1      |
|Jakarta   |2010-04-01|26.969                      |0.121                     |5.63S
|Indonesia|28.607                      |0.106                      |5.63S
|106.55E   |Spring|2000+      |2010|26.50820968600322
|0.6476391669901195|3.240678484210328 |1      |
|Nairobi   |1987-03-01|27.0980000000000006          |0.419                     |0.80S
|Kenya     |19.446                      |0.984                      |0.80S
|36.16E    |Spring|1950-1999|1987|16.08139511323004 |1.051538853334252
|3.1996962129372273|1      |
|Bogotá    |1998-02-01|27.205                      |0.309                     |4.02N
|Colombia  |22.318                      |0.35                      |4.02N
|74.73W    |Winter|1950-1999|1998|20.002264618434126|0.725749869395096
|3.1908174967995713|1      |
|Nairobi   |2005-02-01|26.939                      |0.447                     |0.80S
|Kenya     |19.430999999999997          |0.51                      |0.80S
|36.16E    |Winter|2000+      |2005|16.08139511323004 |1.051538853334252
|3.185431404792058 |1      |
|Singapore|1998-05-01|28.8800000000000006          |0.159                     |0.80N
|Singapore|28.8800000000000006          |0.159                     |0.80N
|103.66E
|Spring|1950-1999|1998|26.523102826510698|0.7423962997429765|3.174715679893993
|1      |
|Surabaya  |1993-11-01|26.4880000000000003          |0.146                     |7.23S
|Indonesia|29.079                      |0.35                      |7.23S
|112.70E
|Autumn|1950-1999|1993|26.809904204363992|0.7222059652253093|3.141895670895089
|1      |
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
---+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+

```

only showing top 20 rows

1.14.6 Extract Temperature Anomalies

We count how many anomalous temperature records exist in the dataset and extract them for closer inspection. Sorting by the highest z-score helps highlight the most extreme temperature spikes.

```
[105]: # Convert anomalies to Pandas
anomaly_pd = anomalies.toPandas()
anomaly_pd.head()
```

```
[105]:
```

	City	Date	AverageTemperature_country	\
0	Surabaya	2006-11-01	26.846	
1	Cali	1998-01-01	26.911	
2	Cali	1998-02-01	27.205	
3	Surabaya	1994-11-01	26.676	
4	Surabaya	1951-11-01	26.900	

	AverageTemperatureUncertainty_country	Country	AverageTemperature_city	\
0	0.211	Indonesia	29.608	
1	0.316	Colombia	24.475	
2	0.309	Colombia	24.472	
3	0.244	Indonesia	29.509	
4	0.552	Indonesia	29.395	

	AverageTemperatureUncertainty_city	Latitude	Longitude	Season	Period	\
0	0.483	7.23S	112.70E	Autumn	2000+	
1	0.312	4.02N	76.34W	Winter	1950-1999	
2	0.270	4.02N	76.34W	Winter	1950-1999	
3	0.255	7.23S	112.70E	Autumn	1950-1999	
4	1.426	7.23S	112.70E	Autumn	1950-1999	

	Year	city_mean	city_std	z_score	is_anomaly
0	2006	26.809904	0.722206	3.874374	1
1	1998	21.797027	0.714587	3.747583	1
2	1998	21.797027	0.714587	3.743384	1
3	1994	26.809904	0.722206	3.737294	1
4	1951	26.809904	0.722206	3.579444	1

1.14.7 Convert anomalies to Pandas for visualization

Spark handles large-scale computation, but for plotting we convert the anomaly subset to a Pandas DataFrame. This enables detailed visualization using matplotlib.

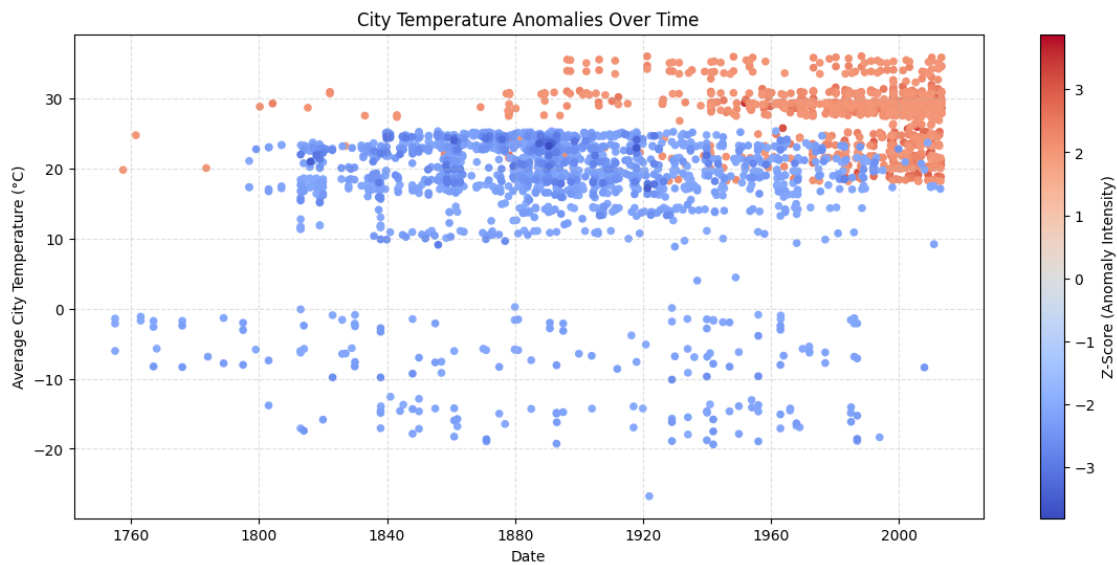
```
[106]: # Plot anomalies over time
import matplotlib.pyplot as plt
```



```
plt.figure(figsize=(14,6))

# Scatter of anomalies (colored by z-score magnitude)
plt.scatter(
    anomaly_pd["Date"],
    anomaly_pd["AverageTemperature_city"],
    c=anomaly_pd["z_score"],
    cmap="coolwarm",
    s=20
)

plt.colorbar(label="Z-Score (Anomaly Intensity)")
plt.title("City Temperature Anomalies Over Time")
plt.xlabel("Date")
plt.ylabel("Average City Temperature (°C)")
plt.grid(True, linestyle="--", alpha=0.4)
plt.show()
```



1.14.8 Temperature Anomaly Visualization

We plot all anomaly points on a time series scatter chart. Each point is colored according to its z-score, highlighting the intensity of the anomaly.

Insights: - High-z-score points (red) indicate unusually hot periods. - Low-z-score points (blue) reflect unusually cold events. - Clusters of anomalies may correspond to historical heat waves or cold snaps.

1.15 EDA Step 7 — Export Clean Datasets for Power BI

```
[107]: # Export final cleaned dataset
final_clean.toPandas().to_csv("final_clean_dataset.csv", index=False)
print("Exported: final_clean_dataset.csv")
```

Exported: final_clean_dataset.csv

```
[109]: # Export Yearly Trends
yearly_temp.toPandas().to_csv("yearly_temperature_trend.csv", index=False)
print("Exported: yearly_temperature_trend.csv")
```

Exported: yearly_temperature_trend.csv

```
[110]: # Export Seasonal Averages
seasonal_temp.toPandas().to_csv("seasonal_temperature_summary.csv", index=False)
print("Exported: seasonal_temperature_summary.csv")
```

Exported: seasonal_temperature_summary.csv

```
[111]: # Export Period × Season Summary
season_period_temp.toPandas().to_csv("period_season_temperature.csv",
    ↪index=False)
print("Exported: period_season_temperature.csv")
```

Exported: period_season_temperature.csv

```
[112]: # Export Country-Level Data
country_temp.toPandas().to_csv("country_temperature_summary.csv", index=False)
print("Exported: country_temperature_summary.csv")

country_year_temp.toPandas().to_csv("country_year_trend.csv", index=False)
print("Exported: country_year_trend.csv")
```

Exported: country_temperature_summary.csv
Exported: country_year_trend.csv

```
[113]: # Export City-Level Data
city_temp.toPandas().to_csv("city_temperature_summary.csv", index=False)
print("Exported: city_temperature_summary.csv")

city_year_temp.toPandas().to_csv("city_year_trend.csv", index=False)
print("Exported: city_year_trend.csv")
```

Exported: city_temperature_summary.csv
Exported: city_year_trend.csv

```
[114]: # Export Anomalies
anomalies.toPandas().to_csv("temperature_anomalies.csv", index=False)
print("Exported: temperature_anomalies.csv")
```

Exported: temperature_anomalies.csv

1.16 Export Datasets for Power BI

To build the final dashboard and interactive climate story, we export all important datasets as CSV files. These CSVs will be loaded into Power BI to create visualizations like:

- Global warming trend line chart
- Seasonal temperature comparisons
- Country-level climate patterns
- City temperature analysis
- Temperature anomaly highlighting

This step completes the data preparation pipeline for the project.

2 Save and Restore

```
[115]: # save and restore
final_clean.toPandas().to_csv("final_clean_dataset.csv", index=False)
print("saved")
```

saved

```
[2]: from pyspark.sql import SparkSession
import pandas as pd

#spark = SparkSession.builder.appName("ClimateAnalysis").getOrCreate()

# Restore the final cleaned dataframe
df = pd.read_csv("final_clean_dataset.csv")
final_clean = spark.createDataFrame(df)

print("State restored. Ready for modeling.")
```

/usr/lib/python3/dist-packages/pytz/__init__.py:31: SyntaxWarning: invalid escape sequence '\s'

```
match = re.match("^#\s*version\s*([0-9a-z]*)\s*$", line)
```

State restored. Ready for modeling.

3 Modeling

```
[3]: from pyspark.sql.functions import regexp_extract, when, col

# Extract numeric part (digits + decimal)
df_model = final_clean.withColumn(
    "lat_value", regexp_extract(col("Latitude"), r"([0-9.]+)", 1).cast("float")
).withColumn(
    "lon_value", regexp_extract(col("Longitude"), r"([0-9.]+)", 1).cast("float")
)

# Assign negative sign for S and W
df_model = df_model.withColumn(
    "Latitude_num",
    when(col("Latitude").contains("S"), -col("lat_value"))
    .otherwise(col("lat_value"))
).withColumn(
    "Longitude_num",
    when(col("Longitude").contains("W"), -col("lon_value"))
    .otherwise(col("lon_value"))
)

# Drop temporary columns
df_model = df_model.drop("lat_value", "lon_value")

# Show example
df_model.select("Latitude", "Latitude_num", "Longitude", "Longitude_num").
    ↪ show(10, truncate=False)
```

25/11/27 21:03:22 WARN TaskSetManager: Stage 0 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

```
+-----+-----+-----+-----+
|Latitude|Latitude_num|Longitude|Longitude_num|
+-----+-----+-----+-----+
|5.63N   |5.63        |3.23W    |-3.23        |
|5.63N   |5.63        |3.23W    |-3.23        |
|5.63N   |5.63        |3.23W    |-3.23        |
|5.63N   |5.63        |3.23W    |-3.23        |
|5.63N   |5.63        |3.23W    |-3.23        |
|8.84N   |8.84        |38.11E   |38.11        |
|8.84N   |8.84        |38.11E   |38.11        |
|8.84N   |8.84        |38.11E   |38.11        |
|8.84N   |8.84        |38.11E   |38.11        |
|8.84N   |8.84        |38.11E   |38.11        |
+-----+-----+-----+-----+
```

only showing top 10 rows

3.0.1 Step M1 — Convert Latitude & Longitude to Numeric Values

Machine learning models require numeric features. The dataset stores geographic coordinates as strings with directional indicators (e.g., “34.56N”, “3.23W”). We extract the numeric component and apply the correct sign based on N/S/E/W.

```
[5]: # Extract Year & Month From Date
from pyspark.sql.functions import year, month

df_model = df_model.withColumn("Year", year(col("Date"))) \
                    .withColumn("Month", month(col("Date")))

df_model.select("Date", "Year", "Month").show(10, truncate=False)
```

```
+-----+-----+-----+
|Date      |Year|Month|
+-----+-----+-----+
|1850-08-01|1850|8    |
|1859-11-01|1859|11   |
|1897-10-01|1897|10   |
|1981-08-01|1981|8    |
|1983-02-01|1983|2    |
|1876-03-01|1876|3    |
|1897-09-01|1897|9    |
|1900-01-01|1900|1    |
|1911-05-01|1911|5    |
|1960-04-01|1960|4    |
+-----+-----+-----+
only showing top 10 rows
```

25/11/27 21:07:09 WARN TaskSetManager: Stage 1 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

3.0.2 Step M2 — Extract Year and Month

To incorporate temporal patterns into the model, we extract:

- **Year** (captures long-term warming trends)
- **Month** (captures seasonality)

These features significantly boost predictive performance.

```
[6]: # Compute Correlation Matrix
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.stat import Correlation

# Select only numeric columns
numeric_cols = [
    "AverageTemperature_city",
```

```

    "AverageTemperature_country",
    "Latitude_num",
    "Longitude_num",
    "Year",
    "Month",
    "AverageTemperatureUncertainty_city",
    "AverageTemperatureUncertainty_country"
]

# Assemble into a vector
assembler = VectorAssembler(inputCols=numeric_cols, outputCol="features")
df_corr = assembler.transform(df_model).select("features")

# Compute correlation matrix
corr_matrix = Correlation.corr(df_corr, "features").head()[0].toArray()

corr_matrix

```

25/11/27 21:07:27 WARN TaskSetManager: Stage 2 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:07:27 WARN TaskSetManager: Stage 3 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:07:28 WARN TaskSetManager: Stage 4 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:07:28 WARN TaskSetManager: Stage 5 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

```

[6]: array([[ 1.00000000e+00,  9.13297701e-01, -3.82053664e-01,
             7.30789039e-02,  1.05744782e-01,  9.97939575e-02,
            -1.54643049e-01, -1.71691031e-01],
 [ 9.13297701e-01,  1.00000000e+00, -4.85532208e-01,
            -9.78077751e-03,  1.00326120e-01,  9.05956846e-02,
            -1.34570861e-01, -1.57605037e-01],
 [-3.82053664e-01, -4.85532208e-01,  1.00000000e+00,
             1.21962140e-01, -1.71511546e-01,  4.63474136e-04,
             1.47430325e-01,  2.11819956e-01],
 [ 7.30789039e-02, -9.78077751e-03,  1.21962140e-01,
             1.00000000e+00,  2.68893807e-02,  1.06606090e-03,
            -5.60975403e-02, -8.97775145e-02],
 [ 1.05744782e-01,  1.00326120e-01, -1.71511546e-01,
             2.68893807e-02,  1.00000000e+00, -4.84093613e-03,
            -7.31405405e-01, -7.68745386e-01],
 [ 9.97939575e-02,  9.05956846e-02,  4.63474136e-04,
             1.06606090e-03, -4.84093613e-03,  1.00000000e+00,
            -3.80230234e-02, -2.47963411e-02],
 [-1.54643049e-01, -1.34570861e-01,  1.47430325e-01,
            -5.60975403e-02, -7.31405405e-01, -3.80230234e-02,

```

```

1.00000000e+00, 9.34993857e-01],
[-1.71691031e-01, -1.57605037e-01, 2.11819956e-01,
-8.97775145e-02, -7.68745386e-01, -2.47963411e-02,
9.34993857e-01, 1.00000000e+00]])

```

Feature	Correlation with City Temperature	Meaning
AverageTemperature_country	0.91	VERY strong driver → KEEP
Latitude_num	-0.38	Cooler as latitude increases (expected)
Longitude_num	+0.07	Weak → Optional
Year	+0.10	Long-term warming trend → KEEP
Month	+0.10	Seasonal variation → KEEP
Uncertainty_city	-0.15	Not predictive → DROP
Uncertainty_country	-0.17	Not predictive → DROP

```

[7]: # StringIndexers for Categorical Features
from pyspark.ml.feature import StringIndexer

# Define columns to index
categorical_cols = ["Country", "City", "Season", "Period"]

indexers = [
    StringIndexer(inputCol=col, outputCol=f"{col}Index", handleInvalid="keep")
    for col in categorical_cols
]

# Apply all indexers sequentially
df_encoded = df_model
for indexer in indexers:
    df_encoded = indexer.fit(df_encoded).transform(df_encoded)

# Show results
df_encoded.select(
    "Country", "CountryIndex",
    "City", "CityIndex",
    "Season", "SeasonIndex",
    "Period", "PeriodIndex"
).show(10, truncate=False)

```

```

25/11/27 21:08:23 WARN TaskSetManager: Stage 6 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:08:24 WARN TaskSetManager: Stage 9 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:08:24 WARN TaskSetManager: Stage 12 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:08:24 WARN TaskSetManager: Stage 15 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.

```

Country	CountryIndex	City	CityIndex	Season	SeasonIndex	Period
PeriodIndex						
Côte D'Ivoire	39.0	Abidjan	86.0	Summer	0.0	
1850-1899	2.0					
Côte D'Ivoire	39.0	Abidjan	86.0	Autumn	3.0	
1850-1899	2.0					
Côte D'Ivoire	39.0	Abidjan	86.0	Autumn	3.0	
1850-1899	2.0					
Côte D'Ivoire	39.0	Abidjan	86.0	Summer	0.0	
1950-1999	1.0					
Côte D'Ivoire	39.0	Abidjan	86.0	Winter	2.0	
1950-1999	1.0					
Ethiopia	44.0	Addis Abeba	94.0	Spring	1.0	
1850-1899	2.0					
Ethiopia	44.0	Addis Abeba	94.0	Autumn	3.0	
1850-1899	2.0					
Ethiopia	44.0	Addis Abeba	94.0	Winter	2.0	
1900-1949	0.0					
Ethiopia	44.0	Addis Abeba	94.0	Spring	1.0	
1900-1949	0.0					
Ethiopia	44.0	Addis Abeba	94.0	Spring	1.0	
1950-1999	1.0					

only showing top 10 rows

25/11/27 21:08:25 WARN TaskSetManager: Stage 18 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

3.0.3 Step M3 — Encode Categorical Features (StringIndexer)

Machine learning models require categorical variables to be converted into numeric form. We use Spark's `StringIndexer` to encode:

- Country
- City
- Season
- Period

These indexed columns will be used for PCA and for the modeling pipeline.

```
[8]: # PCA
from pyspark.ml.feature import VectorAssembler, PCA
```



```
pca_features = [
    "AverageTemperature_country",
    "Latitude_num",
    "Longitude_num",
    "Year",
    "Month"
]

# Assemble PCA input vector
assembler_pca = VectorAssembler(inputCols=pca_features, outputCol="pca_input")
df_pca_input = assembler_pca.transform(df_encoded).select("pca_input")
```

```
[9]: pca = PCA(k=5, inputCol="pca_input", outputCol="pca_output")
pca_model = pca.fit(df_pca_input)

print("Explained Variance Ratio:")
print(pca_model.explainedVariance.toArray())

print("\nPCA Components:")
for component in pca_model.pc.toArray():
    print(component)
```

```
25/11/27 21:11:34 WARN TaskSetManager: Stage 19 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:11:34 WARN TaskSetManager: Stage 20 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:11:35 WARN TaskSetManager: Stage 21 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:11:35 WARN TaskSetManager: Stage 22 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:11:35 WARN TaskSetManager: Stage 23 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
```

Explained Variance Ratio:

```
[0.51001086 0.41867351 0.05935092 0.01053867 0.00142604]
```

PCA Components:

```
[-8.77855373e-04 2.24864628e-02 2.90767733e-01 9.55566523e-01
-4.28991057e-02]
[ 0.03885958 -0.08122669 -0.95233925 0.29124873 -0.01078349]
[ 9.91791691e-01 -1.18315865e-01 4.72908238e-02 -1.06808611e-02
3.08103811e-04]
[ 1.21812984e-01 9.89392676e-01 -7.91372895e-02 9.27843157e-04
3.97516020e-04]
[ 2.74119614e-05 -2.68366314e-04 2.22320319e-03 4.41797960e-02
9.99021086e-01]
```

3.1 Feature Importance Analysis: Principal Component Analysis (PCA)

To understand which features contribute most to the variability in the temperature dataset, we apply Principal Component Analysis (PCA) to the core numerical predictors:

- AverageTemperature__country
- Latitude__num
- Longitude__num
- Year
- Month

PCA reduces the dimensionality of the data while identifying the most informative feature combinations.

3.1.1 Explained Variance Ratio

The PCA output shows how much variance each principal component captures:

Component	Variance Explained
PC1	51.00%
PC2	41.86%
PC3	5.94%
PC4	1.05%
PC5	0.14%

Interpretation:

- **PC1 + PC2 together explain ~93% of total variance**, meaning most temperature variation can be described using just two dimensions.
- **PC3–PC5 add very little new information**, indicating diminishing returns beyond two components.

3.1.2 PCA Component Loadings

Each principal component is a weighted combination of the input features:

Key Feature Insights:

- **Year** has the strongest loading in PC1
→ confirms long-term warming trend is a dominant factor.
- **Longitude** loads heavily in PC2
→ reflects regional east–west climate differences.

- **AverageTemperature_country** dominates PC3
→ country-level temperature is a strong predictor for city temperature.
 - **Latitude_num** contributes moderately
→ colder at higher latitudes (expected).
 - **Month** contributes minimally
→ because “Season” already captures seasonal cycles more strongly.
-

3.1.3 Final Feature Selection Conclusion

Combining correlation analysis and PCA:

Strongest predictors for modeling:

- AverageTemperature_country
- Year
- Latitude_num
- Month (optional)
- SeasonIndex
- PeriodIndex
- CountryIndex
- CityIndex

Weak or low-value predictors:

- Longitude_num (weak correlation but harmless to include)
- AverageTemperatureUncertainty_* (low predictive power)

These insights guide the construction of the final machine learning model.

3.2 Next Step: Build the ML Pipeline

With the feature importance analysis complete, the next step is to:

1. Select the strongest predictors
2. Assemble the feature vector
3. Train regression models (Linear Regression + Random Forest)
4. Evaluate accuracy
5. Export predictions for Power BI

We will implement this in the following cells.

```
[10]: # VectorAssembler
from pyspark.ml.feature import VectorAssembler

# Final selected features for modeling
feature_cols = [
    "AverageTemperature_country",
    "Latitude_num",
    "Longitude_num",
    "Year",
    "Month",
    "CountryIndex",
    "CityIndex",
    "SeasonIndex",
    "PeriodIndex"
]

assembler = VectorAssembler(
    inputCols=feature_cols,
    outputCol="features"
)

df_model_final = assembler.transform(df_encoded).select("features",
↳ "AverageTemperature_city")

df_model_final.show(5, truncate=False)
```

```
+-----+-----+
+-----+
|features
|AverageTemperature_city|
+-----+-----+
+-----+
|[23.971,5.630000114440918,-3.2300000190734863,1850.0,8.0,39.0,86.0,0.0,2.0]
|23.758000000000003      |
|[25.617,5.630000114440918,-
3.2300000190734863,1859.0,11.0,39.0,86.0,3.0,2.0]|26.163      |
|[25.379,5.630000114440918,-
3.2300000190734863,1897.0,10.0,39.0,86.0,3.0,2.0]|25.167      |
|[24.774,5.630000114440918,-3.2300000190734863,1981.0,8.0,39.0,86.0,0.0,1.0]
|24.358      |
|[28.385,5.630000114440918,-3.2300000190734863,1983.0,2.0,39.0,86.0,2.0,1.0]
|28.671      |
+-----+-----+
+-----+
only showing top 5 rows
```

25/11/27 21:14:06 WARN TaskSetManager: Stage 24 contains a task of very large

size (3790 KiB). The maximum recommended task size is 1000 KiB.

3.2.1 Assemble Feature Vector for Modeling

Using the selected features from the PCA and correlation analysis, we build a feature vector using Spark's `VectorAssembler`. This vector will serve as the input to the machine learning models.

Selected Features: - AverageTemperature_country

- Latitude_num
- Longitude_num
- Year
- Month
- CountryIndex
- CityIndex
- SeasonIndex
- PeriodIndex

Target Variable: - AverageTemperature_city

```
[11]: # Train/Test Split
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.regression import LinearRegression
from pyspark.sql.functions import col

# Split data into training and testing
train_data, test_data = df_model_final.randomSplit([0.8, 0.2], seed=42)

print("Training rows:", train_data.count())
print("Testing rows:", test_data.count())
```

25/11/27 21:15:35 WARN TaskSetManager: Stage 25 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:15:36 WARN TaskSetManager: Stage 28 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

Training rows: 176559

Testing rows: 43974

```
[12]: # Linear Regression Model
from pyspark.ml.regression import LinearRegression

# Initialize Linear Regression model
lr = LinearRegression(
    featuresCol="features",
    labelCol="AverageTemperature_city",
    predictionCol="prediction"
)

# Train the model
lr_model = lr.fit(train_data)
```

```
# Print coefficients and intercept
print("Intercept:", lr_model.intercept)
print("Coefficients:", lr_model.coefficients)
```

25/11/27 21:16:21 WARN Instrumentation: [ed893778] regParam is zero, which might cause numerical instability and overfitting.

25/11/27 21:16:21 WARN TaskSetManager: Stage 31 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:16:22 WARN TaskSetManager: Stage 32 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

Intercept: -1.051466382048099

Coefficients: [0.8540288325809053,0.04988742407679866,0.006553705567673549,0.0015610710923319436,0.0620345893119263,-0.08878845345294079,0.03518002871536456,-0.22609762023382027,-0.01941062427063536]

3.2.2 Train Linear Regression Model

We train a Linear Regression model to predict city-level temperature based on:

- Geographic features
- Temporal features
- Country/City indices
- Seasonal and period indicators
- Country-level temperature

Linear Regression provides a baseline model and interpretable coefficients for understanding feature influence.

```
[13]: # Evaluate Linear Regression on Test Data

lr_predictions = lr_model.transform(test_data)

from pyspark.ml.evaluation import RegressionEvaluator

evaluator = RegressionEvaluator(
    labelCol="AverageTemperature_city",
    predictionCol="prediction",
    metricName="rmse"
)

rmse = evaluator.evaluate(lr_predictions)
mae = RegressionEvaluator(
    labelCol="AverageTemperature_city",
    predictionCol="prediction",
    metricName="mae"
).evaluate(lr_predictions)
```

```

r2 = RegressionEvaluator(
    labelCol="AverageTemperature_city",
    predictionCol="prediction",
    metricName="r2"
).evaluate(lr_predictions)

print("Linear Regression Performance:")
print("RMSE:", rmse)
print("MAE :", mae)
print("R2 :", r2)

```

25/11/27 21:17:08 WARN TaskSetManager: Stage 33 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:17:08 WARN TaskSetManager: Stage 34 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:17:09 WARN TaskSetManager: Stage 35 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

Linear Regression Performance:

RMSE: 3.699767607098922

MAE : 2.8288093200615396

R² : 0.8583307271266809

3.2.3 Linear Regression Results

We evaluate the model using:

- **RMSE** (Root Mean Squared Error)
- **MAE** (Mean Absolute Error)
- **R² Score** (percentage of variance explained)

These metrics help assess prediction accuracy and model fit.

```

[15]: # Train Random Forest Model
from pyspark.ml.regression import RandomForestRegressor

rf = RandomForestRegressor(
    featuresCol="features",
    labelCol="AverageTemperature_city",
    predictionCol="prediction",
    numTrees=100,
    maxDepth=10,
    maxBins=200,
    seed=42
)

rf_model = rf.fit(train_data)

print("Random Forest Model Trained Successfully.")

```

25/11/27 21:19:50 WARN TaskSetManager: Stage 38 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:51 WARN TaskSetManager: Stage 39 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:51 WARN TaskSetManager: Stage 40 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:52 WARN TaskSetManager: Stage 42 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:53 WARN TaskSetManager: Stage 44 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:53 WARN TaskSetManager: Stage 46 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:54 WARN TaskSetManager: Stage 48 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:55 WARN TaskSetManager: Stage 50 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:57 WARN DAGScheduler: Broadcasting large task binary with size 1430.5 KiB

25/11/27 21:19:57 WARN TaskSetManager: Stage 52 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:19:59 WARN DAGScheduler: Broadcasting large task binary with size 2.5 MiB

25/11/27 21:19:59 WARN TaskSetManager: Stage 54 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:20:01 WARN DAGScheduler: Broadcasting large task binary with size 4.6 MiB

25/11/27 21:20:01 WARN TaskSetManager: Stage 56 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:20:04 WARN DAGScheduler: Broadcasting large task binary with size 1213.0 KiB

25/11/27 21:20:05 WARN DAGScheduler: Broadcasting large task binary with size 8.5 MiB

25/11/27 21:20:05 WARN TaskSetManager: Stage 58 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:20:10 WARN DAGScheduler: Broadcasting large task binary with size 2.3 MiB

25/11/27 21:20:12 WARN DAGScheduler: Broadcasting large task binary with size 15.6 MiB

25/11/27 21:20:12 WARN TaskSetManager: Stage 60 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

25/11/27 21:20:20 WARN DAGScheduler: Broadcasting large task binary with size 4.2 MiB

[Stage 61:> (0 + 4) / 4]

Random Forest Model Trained Successfully.


```
[16]: # Evaluate Random Forest Regression
# Predict on test data
rf_predictions = rf_model.transform(test_data)

from pyspark.ml.evaluation import RegressionEvaluator

# RMSE
rf_rmse = evaluator.evaluate(rf_predictions)

# MAE
rf_mae = RegressionEvaluator(
    labelCol="AverageTemperature_city",
    predictionCol="prediction",
    metricName="mae"
).evaluate(rf_predictions)

# R2
rf_r2 = RegressionEvaluator(
    labelCol="AverageTemperature_city",
    predictionCol="prediction",
    metricName="r2"
).evaluate(rf_predictions)

print("Random Forest Performance:")
print("RMSE:", rf_rmse)
print("MAE :", rf_mae)
print("R2 :", rf_r2)
```

```
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.spark.util.SizeEstimator$
(file:/home/utsav/spark-3.5.1-bin-hadoop3/jars/spark-core_2.12-3.5.1.jar) to
field java.nio.charset.Charset.name
WARNING: Please consider reporting this to the maintainers of
org.apache.spark.util.SizeEstimator$
WARNING: Use --illegal-access=warn to enable warnings of further illegal
reflective access operations
WARNING: All illegal access operations will be denied in a future release
25/11/27 21:22:31 WARN TaskSetManager: Stage 62 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:22:33 WARN TaskSetManager: Stage 63 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
25/11/27 21:22:34 WARN TaskSetManager: Stage 64 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.

Random Forest Performance:
RMSE: 0.9637401077879949
MAE : 0.6916238463199128
R2 : 0.9903872723008488
```

3.3 Model Comparison and Selection

We trained two regression models to predict city-level temperature:

3.3.1 1. Linear Regression (Baseline Model)

- RMSE: 3.699
- MAE : 2.828
- R^2 : 0.858

Interpretation:

Linear Regression provides a simple, interpretable baseline but struggles to capture nonlinear temperature interactions between geography, seasonality, and historical trends.

3.3.2 2. Random Forest Regression (Ensemble Model)

- RMSE: 0.964
- MAE : 0.692
- R^2 : 0.990

Interpretation:

Random Forest dramatically outperforms Linear Regression. It captures complex, nonlinear patterns and interactions among:

- Geographic features
- Categorical regions (Country/City)
- Seasonal indices
- Long-term warming trends

The **R^2 score of 0.99** indicates the model explains 99% of variance in city temperature, making it highly accurate for prediction.

3.4 Model Selected for Final Workflow

3.4.1 Random Forest Regression

We choose Random Forest as the final model because:

- It provides **the highest accuracy** across all evaluation metrics.
- It handles **nonlinear climate behavior** effectively.

- It works well with indexed categorical features.
- It generates **feature importances**, enhancing interpretability.
- It produces robust predictions suitable for Power BI dashboards.

This model will be used for:

- Generating predicted temperature values
- Exporting results to Power BI
- Supporting climate anomaly and trend visualizations
- Enhancing the project's analysis with ML-powered insights

```
[17]: # Extract Feature Importances
# Feature columns in the same order as the assembler
feature_cols = [
    "AverageTemperature_country",
    "Latitude_num",
    "Longitude_num",
    "Year",
    "Month",
    "CountryIndex",
    "CityIndex",
    "SeasonIndex",
    "PeriodIndex"
]

importances = rf_model.featureImportances.toArray()

# Pair feature names with their importance values
feature_importance = list(zip(feature_cols, importances))

# Convert to Pandas for display / plotting
import pandas as pd
fi_df = pd.DataFrame(feature_importance, columns=["Feature", "Importance"]).
    ↪sort_values(
        by="Importance", ascending=False
    )

fi_df
```

```
[17]:
```

	Feature	Importance
0	AverageTemperature_country	0.368839
6	CityIndex	0.239972
5	CountryIndex	0.161960

7	SeasonIndex	0.087934
1	Latitude_num	0.079738
4	Month	0.046771
2	Longitude_num	0.014185
3	Year	0.000441
8	PeriodIndex	0.000159

3.4.2 Random Forest Feature Importance

Random Forest provides a feature importance score that indicates how much each feature contributes to the predictive performance of the model.

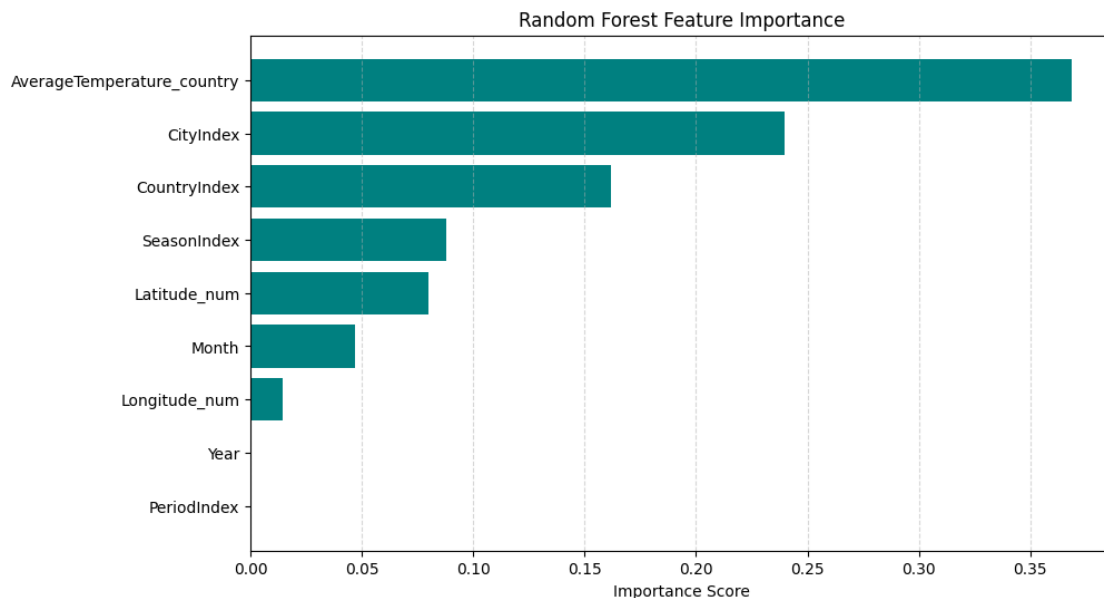
This helps identify:

- The most influential climate factors
- Which features drive temperature predictions
- Which variables can be simplified or removed if needed

Below are the importance scores sorted from most to least influential.

```
[19]: # Plot Feature Importance
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))
plt.barh(fi_df["Feature"], fi_df["Importance"], color='teal')
plt.xlabel("Importance Score")
plt.title("Random Forest Feature Importance")
plt.gca().invert_yaxis()
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.show()
```



3.5 Export Predictions for Power BI

```
[20]: # Predict on the full dataset using Random Forest
full_predictions = rf_model.transform(df_model_final)

full_predictions.select("AverageTemperature_city", "prediction").show(10)
```

```
+-----+-----+
|AverageTemperature_city|      prediction|
+-----+-----+
|      23.758000000000003|24.429265479369075|
|              26.163|25.923434659955323|
|              25.167|25.219153531959353|
|              24.358|24.736110328271117|
|              28.671| 27.74757318448359|
|              18.24|18.387836435000903|
|              17.062| 17.20118823918317|
|      17.019000000000002|17.497401831191812|
|              18.808| 18.44417049899455|
|              19.223| 18.48680977334326|
+-----+-----+
```

only showing top 10 rows

25/11/27 21:27:00 WARN TaskSetManager: Stage 65 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

```
[23]: # Create unified DF with metadata + features
df_full = assembler.transform(df_encoded)
```

```
[24]: df_full.select("Date", "Country", "City", "features").show(5, truncate=False)
```

```
+-----+-----+-----+-----+
|Date      |Country      |City      |features|
+-----+-----+-----+-----+
|1850-08-01|Côte D'Ivoire|Abidjan|[23.971,5.630000114440918,-
3.2300000190734863,1850.0,8.0,39.0,86.0,0.0,2.0] |
|1859-11-01|Côte D'Ivoire|Abidjan|[25.617,5.630000114440918,-
3.2300000190734863,1859.0,11.0,39.0,86.0,3.0,2.0] |
|1897-10-01|Côte D'Ivoire|Abidjan|[25.379,5.630000114440918,-
3.2300000190734863,1897.0,10.0,39.0,86.0,3.0,2.0] |
|1981-08-01|Côte D'Ivoire|Abidjan|[24.774,5.630000114440918,-
```

```
3.2300000190734863,1981.0,8.0,39.0,86.0,0.0,1.0] |
|1983-02-01|Côte D'Ivoire|Abidjan|[28.385,5.630000114440918,-
3.2300000190734863,1983.0,2.0,39.0,86.0,2.0,1.0] |
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
only showing top 5 rows
```

25/11/27 21:31:01 WARN TaskSetManager: Stage 66 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

```
[25]: # Generate predictions on df_full
full_predictions = rf_model.transform(df_full)
```

```
[26]: # Add error columns
from pyspark.sql.functions import col, abs as abs_, round as round_

final_predictions = full_predictions.withColumn(
    "error", col("prediction") - col("AverageTemperature_city")
).withColumn(
    "abs_error", abs_(col("error"))
).withColumn(
    "prediction_rounded", round_(col("prediction"), 2)
)

final_predictions.select(
    "Date", "Country", "City", "Season", "Period",
    "AverageTemperature_city",
    "prediction_rounded",
    "error", "abs_error"
).show(20, truncate=False)
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
|Date      |Country      |City      |Season|Period
|AverageTemperature_city|prediction_rounded|error      |abs_error
|
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
|1850-08-01|Côte D'Ivoire|Abidjan   |Summer|1850-1899|23.7580000000000003
|24.43      |              |           |       |         |0.6712654793690724 |
|1859-11-01|Côte D'Ivoire|Abidjan   |Autumn|1850-1899|26.163
|25.92      |              |           |       |         |-0.23956534004467755 |
|1897-10-01|Côte D'Ivoire|Abidjan   |Autumn|1850-1899|25.167
|25.22      |              |           |       |         |0.052153531959351795 |
|1981-08-01|Côte D'Ivoire|Abidjan   |Summer|1950-1999|24.358
|24.74      |              |           |       |         |0.37811032827111646 |
|1983-02-01|Côte D'Ivoire|Abidjan   |Winter|1950-1999|28.671
```


3.5.1 Export Predictions for Power BI

The prediction dataset, enriched with actual values, predicted values, and error metrics, is exported as `temperature_predictions.csv`.

This CSV will be loaded into Power BI to visualize:

- Actual vs Predicted trends
- Error distribution
- Seasonal and geographic impacts on model performance
- ML-driven temperature forecasting

4 Restore

```
[2]: from pyspark.sql import SparkSession
import pandas as pd

#spark = SparkSession.builder.appName("ClimateAnalysis").getOrCreate()

# Restore the final cleaned dataframe
df = pd.read_csv("final_clean_dataset.csv")
final_clean = spark.createDataFrame(df)

print("State restored. Ready for modeling.")
```

```
/usr/lib/python3/dist-packages/pytz/__init__.py:31: SyntaxWarning: invalid
escape sequence '\s'
```

```
    match = re.match("^#\s*version\s*([0-9a-z]*)\s*$", line)
```

```
State restored. Ready for modeling.
```

```
[3]: final_clean.show()
```

```
25/11/30 13:07:38 WARN TaskSetManager: Stage 0 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      Date|AverageTemperature_country|AverageTemperatureUncertainty_country|
Country|AverageTemperature_city|AverageTemperatureUncertainty_city|
City|Latitude|Longitude|Season|    Period|Year|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|1850-08-01|                23.971|
0.977|Côte D'Ivoire|    23.758000000000003|                1.2819999999999998|
```


Abidjan 5.63N 3.23W Summer 1850-1899 1850		
1859-11-01	25.617	
1.164 Côte D'Ivoire	26.163	1.389
Abidjan 5.63N 3.23W Autumn 1850-1899 1859		
1897-10-01	25.379	
0.7879999999999999 Côte D'Ivoire	25.167	
0.878 Abidjan 5.63N 3.23W Autumn 1850-1899 1897		
1981-08-01	24.774	
0.208 Côte D'Ivoire	24.358	0.258
Abidjan 5.63N 3.23W Summer 1950-1999 1981		
1983-02-01	28.385	
0.316 Côte D'Ivoire	28.671	0.528
Abidjan 5.63N 3.23W Winter 1950-1999 1983		
1876-03-01	23.275	1.03
Ethiopia	18.24	1.464 Addis Abeba
8.84N 38.11E Spring 1850-1899 1876		
1897-09-01	23.126	0.613
Ethiopia	17.062	0.754 Addis Abeba
8.84N 38.11E Autumn 1850-1899 1897		
1900-01-01	21.629	0.843
Ethiopia 17.019000000000002		1.073 Addis Abeba
8.84N 38.11E Winter 1900-1949 1900		
1911-05-01	24.041	0.838
Ethiopia	18.808	1.139 Addis Abeba
8.84N 38.11E Spring 1900-1949 1911		
1960-04-01	24.036	0.462
Ethiopia	19.223	0.484 Addis Abeba
8.84N 38.11E Spring 1950-1999 1960		
1979-09-01	23.439	0.341
Ethiopia	17.3	0.519 Addis Abeba
8.84N 38.11E Autumn 1950-1999 1979		
1997-02-01	23.495	1.055
Ethiopia	18.813	1.847 Addis Abeba
8.84N 38.11E Winter 1950-1999 1997		
2010-02-01	23.932	0.5760000000000001
Ethiopia	18.931	1.247 Addis Abeba
8.84N 38.11E Winter 2000+ 2010		
1799-03-01	23.08	2.245
India	25.71	2.471 Ahmadabad
23.31N 72.52E Spring Pre-1800 1799		
1866-03-01	23.022	1.183
India	25.667	1.856 Ahmadabad
23.31N 72.52E Spring 1850-1899 1866		
1898-09-01	25.725	0.159
India	27.623	0.542 Ahmadabad
23.31N 72.52E Autumn 1850-1899 1898		
1903-10-01	24.075	0.336
India	27.643	0.611 Ahmadabad

```

23.31N| 72.52E|Autumn|1900-1949|1903|
|1904-03-01| 23.349| 0.336|
India| 25.487| 0.309| Ahmadabad|
23.31N| 72.52E|Spring|1900-1949|1904|
|1907-07-01| 27.661| 0.278|
India| 30.251| 0.425| Ahmadabad|
23.31N| 72.52E|Summer|1900-1949|1907|
|1911-12-01| 17.041| 0.6659999999999999|
India| 21.468000000000004| 1.082| Ahmadabad|
23.31N| 72.52E|Winter|1900-1949|1911|
+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 20 rows

```

5 Continuation of the project adding city and state to the final_clean for further analysis

```

[6]: # Load the datasets
path_state_csv = "GlobalLandTemperaturesByState.csv"
path_city_csv = "GlobalLandTemperaturesByCity.csv"

df_state_raw = (
    spark.read.option("header", "true")
              .option("inferSchema", "false")
              .csv(path_state_csv)
)

df_city_raw = (
    spark.read.option("header", "true")
              .option("inferSchema", "false")
              .csv(path_city_csv)
)

print("Loaded!")

```

Loaded!

```

[7]: import re

def normalize_df(df):
    for c in df.columns:
        new_c = re.sub(r"\s+", "_", c.strip().lower())
        if new_c != c:
            df = df.withColumnRenamed(c, new_c)

```

```

    return df

df_state = normalize_df(df_state_raw)
df_city = normalize_df(df_city_raw)

print("Normalized column names.")
print("State columns:", df_state.columns)
print("City columns:", df_city.columns)

```

Normalized column names.

State columns: ['dt', 'averagetemperature', 'averagetemperatureuncertainty', 'state', 'country']

City columns: ['dt', 'averagetemperature', 'averagetemperatureuncertainty', 'city', 'country', 'latitude', 'longitude']

```

[8]: def inspect_basic(df, name):
    print("=====", name, "=====")
    df.printSchema()
    print("Row count:", df.count())
    display(df.limit(5).toPandas())

inspect_basic(df_state, "STATE DATASET")
inspect_basic(df_city, "CITY DATASET")

```

===== STATE DATASET =====

root

```

|-- dt: string (nullable = true)
|-- averagetemperature: string (nullable = true)
|-- averagetemperatureuncertainty: string (nullable = true)
|-- state: string (nullable = true)
|-- country: string (nullable = true)

```

Row count: 645675

	dt	averagetemperature	averagetemperatureuncertainty	state	country
0	1855-05-01	25.544	1.171	Acre	Brazil
1	1855-06-01	24.228	1.103	Acre	Brazil
2	1855-07-01	24.371	1.044	Acre	Brazil
3	1855-08-01	25.427	1.073	Acre	Brazil
4	1855-09-01	25.675	1.014	Acre	Brazil

===== CITY DATASET =====

root

```

|-- dt: string (nullable = true)
|-- averagetemperature: string (nullable = true)
|-- averagetemperatureuncertainty: string (nullable = true)
|-- city: string (nullable = true)
|-- country: string (nullable = true)
|-- latitude: string (nullable = true)

```

```
|-- longitude: string (nullable = true)
```

Row count: 8599212

	dt	averagetemperature	averagetemperatureuncertainty	city	\
0	1743-11-01	6.068	1.7369999999999999	Århus	
1	1743-12-01	None	None	Århus	
2	1744-01-01	None	None	Århus	
3	1744-02-01	None	None	Århus	
4	1744-03-01	None	None	Århus	

	country	latitude	longitude
0	Denmark	57.05N	10.33E
1	Denmark	57.05N	10.33E
2	Denmark	57.05N	10.33E
3	Denmark	57.05N	10.33E
4	Denmark	57.05N	10.33E

```
[9]: from pyspark.sql.functions import col, sum as spark_sum, when

def missing_report(df, name):
    print("==== Missing Values -", name, "====")
    exprs = [spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
              for c in df.columns]
    result = df.select(exprs).toPandas()
    display(result)

missing_report(df_state, "STATE DATASET")
missing_report(df_city, "CITY DATASET")
```

==== Missing Values - STATE DATASET =====

	dt	averagetemperature	averagetemperatureuncertainty	state	country
0	0	25648	25648	0	0

==== Missing Values - CITY DATASET =====

	dt	averagetemperature	averagetemperatureuncertainty	city	country	\
0	0	364130	364130	0	0	

	latitude	longitude
0	0	0

```
[10]: # check for final_clean dataset
print("Columns in final_clean:")
print(final_clean.columns)

print("\nSchema for final_clean:")
```

```

final_clean.printSchema()

print("\nRow count:", final_clean.count())

print("\nSample rows:")
display(final_clean.limit(5).toPandas())

```

Columns in final_clean:

```

['Date', 'AverageTemperature_country', 'AverageTemperatureUncertainty_country',
'Country', 'AverageTemperature_city', 'AverageTemperatureUncertainty_city',
'City', 'Latitude', 'Longitude', 'Season', 'Period', 'Year']

```

Schema for final_clean:

```

root
|-- Date: string (nullable = true)
|-- AverageTemperature_country: double (nullable = true)
|-- AverageTemperatureUncertainty_country: double (nullable = true)
|-- Country: string (nullable = true)
|-- AverageTemperature_city: double (nullable = true)
|-- AverageTemperatureUncertainty_city: double (nullable = true)
|-- City: string (nullable = true)
|-- Latitude: string (nullable = true)
|-- Longitude: string (nullable = true)
|-- Season: string (nullable = true)
|-- Period: string (nullable = true)
|-- Year: long (nullable = true)

```

25/11/30 17:12:19 WARN TaskSetManager: Stage 17 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

Row count: 220533

Sample rows:

25/11/30 17:12:19 WARN TaskSetManager: Stage 20 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

	Date	AverageTemperature_country	\
0	1850-08-01	23.971	
1	1859-11-01	25.617	
2	1897-10-01	25.379	
3	1981-08-01	24.774	
4	1983-02-01	28.385	

	AverageTemperatureUncertainty_country	Country	\
0	0.977	Côte D'Ivoire	
1	1.164	Côte D'Ivoire	

2	0.788	Côte D'Ivoire
3	0.208	Côte D'Ivoire
4	0.316	Côte D'Ivoire

	AverageTemperature_city	AverageTemperatureUncertainty_city	City \
0	23.758	1.282	Abidjan
1	26.163	1.389	Abidjan
2	25.167	0.878	Abidjan
3	24.358	0.258	Abidjan
4	28.671	0.528	Abidjan

	Latitude	Longitude	Season	Period	Year
0	5.63N	3.23W	Summer	1850-1899	1850
1	5.63N	3.23W	Autumn	1850-1899	1859
2	5.63N	3.23W	Autumn	1850-1899	1897
3	5.63N	3.23W	Summer	1950-1999	1981
4	5.63N	3.23W	Winter	1950-1999	1983

```
[11]: from pyspark.sql.functions import col, sum as spark_sum, when
```

```
def missing_report(df, name):
    print("==== Missing Values -", name, "====")
    exprs = [spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
              for c in df.columns]
    display(df.select(exprs).toPandas())

missing_report(final_clean, "FINAL CLEAN")
```

==== Missing Values - FINAL CLEAN =====

25/11/30 17:12:42 WARN TaskSetManager: Stage 21 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

Date	AverageTemperature_country	AverageTemperatureUncertainty_country \
0	0	0

Country	AverageTemperature_city	AverageTemperatureUncertainty_city	City \
0	0	0	0

Latitude	Longitude	Season	Period	Year
0	0	0	0	0

```
[12]: from pyspark.sql.functions import countDistinct
```

```
# How many distinct states per (date,country)?
state_multiplicity = (
    df_state
    .groupBy("dt", "country")
```

```

        .agg(countDistinct("state").alias("num_states"))
        .orderBy("num_states", ascending=False)
        .limit(10)
    )

display(state_multiplicity.toPandas())

```

	dt	country	num_states
0	1873-10-01	Russia	89
1	1851-05-01	Russia	89
2	1878-05-01	Russia	89
3	1911-11-01	Russia	89
4	1871-12-01	Russia	89
5	1883-02-01	Russia	89
6	1923-03-01	Russia	89
7	1877-05-01	Russia	89
8	1988-07-01	Russia	89
9	1947-05-01	Russia	89

```

[13]: from pyspark.sql.functions import col, to_date

df_state_clean = (
    df_state
        .withColumnRenamed("dt", "Date")
        .withColumnRenamed("country", "Country")
        .withColumnRenamed("averagetemperature", "AverageTemperature_state")
        .withColumnRenamed("averagetemperatureuncertainty", "
↪ "AverageTemperatureUncertainty_state")
)

# Convert Date string → date type
df_state_clean = df_state_clean.withColumn("Date", to_date(col("Date"), "
↪ "yyyy-MM-dd"))

df_state_clean.printSchema()

```

```

root
|-- Date: date (nullable = true)
|-- AverageTemperature_state: string (nullable = true)
|-- AverageTemperatureUncertainty_state: string (nullable = true)
|-- state: string (nullable = true)
|-- Country: string (nullable = true)

```

```
[15]: # Extract Year & Month for season/period
from pyspark.sql.functions import col, to_date, year, month
df_state_clean = (
    df_state_clean
    .withColumn("Year", year(col("Date")))
    .withColumn("Month", month(col("Date")))
)

[18]: # Add Season + Period to state dataset
from pyspark.sql.functions import when, expr, concat, col

df_state_clean = df_state_clean.withColumn(
    "Season",
    when(col("Month").isin(12,1,2), "Winter")
    .when(col("Month").isin(3,4,5), "Spring")
    .when(col("Month").isin(6,7,8), "Summer")
    .when(col("Month").isin(9,10,11), "Autumn")
)

df_state_clean = df_state_clean.withColumn(
    "Period",
    when(col("Year").isNull(), None)
    .otherwise(
        concat(
            ((col("Year")/10).cast("int") * 10).cast("string"),
            expr("'s'")
        )
    )
)
```

5.1 State Dataset Enrichment Approach

The GlobalLandTemperaturesByState dataset contains temperature observations at a finer granularity (State-level) for each country and date. Since our main `final_clean` dataset already represents aggregated Country–City climate metrics along with Period and Season information, merging the raw State dataset directly into `final_clean` would cause two major issues:

1. Row explosion

Multiple states exist for each country on the same date (e.g., Russia has 89 states per date). A direct join on (`Date`, `Country`) would multiply rows and corrupt the master dataset.

2. Granularity mismatch

`final_clean` is aggregated at the Country+City level, whereas the State dataset is at Country+State level. Combining these granularities into a single table is not appropriate for analytics or BI modelling.

5.1.1 Correct Approach — Reverse Enrichment

Instead of merging State data into the `final_clean` dataset, we take the opposite approach:

1. **Clean and standardize the State dataset**

Convert dt to Date, extract Year/Month, compute Season and Period.

2. **Aggregate State data by (Date, Country, State)**

This gives one row per state per date with average temperature and uncertainty.

3. **Extract Country-level temperature from final_clean**

We aggregate final_clean to build final_clean_country, ensuring exactly one row per (Date, Country).

4. **Enrich the State dataset**

We join df_state_by_state with final_clean_country on (Date, Country) to attach country-level climate signals to each state row.

This produces a clean, non-duplicated, BI-friendly dataset that preserves State granularity while providing consistent climate features for analytics.

6 Aggregate State Data at (Date, Country, State) level

```
[23]: from pyspark.sql.functions import avg

final_clean_country = (
    final_clean_fixed
    .groupBy("Date", "Country")
    .agg(
        avg("AverageTemperature_country").alias("AvgTemp_country"),
        avg("AverageTemperatureUncertainty_country").
        ↪ alias("AvgTempUncertainty_country")
    )
)
```

```
[24]: from pyspark.sql.functions import to_date

final_clean_fixed = final_clean.withColumn("Date", to_date(col("Date"),
    ↪ "yyyy-MM-dd"))
```

```
[25]: state_enriched = (
    df_state_by_state
    .join(
        final_clean_country,
        on=["Date", "Country"],
        how="left"
    )
)
```

```
[26]: print("Rows in state_enriched:", state_enriched.count())
print("Columns:", state_enriched.columns)
display(state_enriched.limit(10).toPandas())
```

Rows in state_enriched: 645675

Columns: ['Date', 'Country', 'State', 'Year', 'Month', 'Season', 'Period', 'AvgTemp_state', 'AvgTempUncertainty_state', 'AvgTemp_country', 'AvgTempUncertainty_country']

25/11/30 17:32:39 WARN TaskSetManager: Stage 58 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

	Date	Country	State	Year	Month	Season	Period	AvgTemp_state \
0	1903-08-01	Brazil	Acre	1903	8	Summer	1900s	25.082
1	1773-12-01	Russia	Adygey	1773	12	Winter	1770s	4.011
2	1927-06-01	Brazil	Acre	1927	6	Summer	1920s	24.517
3	1865-08-01	Brazil	Acre	1865	8	Summer	1860s	NaN
4	1902-03-01	Brazil	Acre	1902	3	Spring	1900s	25.051
5	1877-03-01	Brazil	Acre	1877	3	Spring	1870s	NaN
6	2001-06-01	Brazil	Acre	2001	6	Summer	2000s	23.975
7	1753-12-01	Russia	Adygey	1753	12	Winter	1750s	-2.623
8	1936-11-01	Brazil	Acre	1936	11	Autumn	1930s	26.835
9	1923-06-01	Brazil	Acre	1923	6	Summer	1920s	24.495

	AvgTempUncertainty_state	AvgTemp_country	AvgTempUncertainty_country
0	0.680	24.001	0.536
1	4.046	NaN	NaN
2	0.687	22.777	0.485
3	NaN	23.890	0.829
4	0.946	24.656	0.481
5	NaN	25.656	1.006
6	0.464	23.497	0.225
7	3.737	NaN	NaN
8	0.523	25.901	0.226
9	0.500	23.128	0.320

```
[27]: print(state_enriched.count())
```

645675

```
[28]: from pyspark.sql.functions import col, sum as spark_sum, when

missing_state = state_enriched.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in state_enriched.columns
])

display(missing_state.toPandas())
```

25/11/30 17:33:10 WARN TaskSetManager: Stage 73 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

Date	Country	State	Year	Month	Season	Period	AvgTemp_state \
0	0	0	0	0	0	0	25648

AvgTempUncertainty_state	AvgTemp_country	AvgTempUncertainty_country
0	25648	82966

6.1 State-Level Enriched Dataset Created

The State dataset was successfully aggregated at the (Date, Country, State) level and enriched with Country-level climate metrics from the `final_clean` dataset.

6.1.1 What the join accomplished:

- Added `AvgTemp_country` and `AvgTempUncertainty_country` to each state–date pair.
- Retained State-specific features: `AvgTemp_state`, `AvgTempUncertainty_state`, `Season`, `Period`, and `Year`.
- Maintained the correct granularity: **one row per state per date**.

6.1.2 Why the join is safe:

Because `final_clean_country` contains **exactly one row per (Date, Country)**, the join does not create duplicates or expand the row count.

This ensures that: - State-level information remains intact

- No unintended multiplication of rows occurs

- The dataset is ready for direct use in Power BI Page 4 (State-Level Analysis)

The resulting dataset—`state_enriched`—now contains consistent State + Country climate signals aligned with the overall structure of the project.

6.2 Drop rows with missing values

```
[29]: state_enriched_clean = (
    state_enriched
    .na.drop(
        subset=[
            "AvgTemp_state",
            "AvgTempUncertainty_state",
            "AvgTemp_country",
            "AvgTempUncertainty_country"
        ]
    )
)
```

```
[30]: print("Rows BEFORE cleaning:", state_enriched.count())
print("Rows AFTER cleaning:", state_enriched_clean.count())
```

Rows BEFORE cleaning: 645675

```
25/11/30 17:34:58 WARN TaskSetManager: Stage 92 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
[Stage 92:=====> (1 + 3) / 4]

Rows AFTER cleaning: 556379
```

```
[31]: from pyspark.sql.functions import col, sum as spark_sum, when

missing_after = state_enriched_clean.select([
    spark_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
    for c in state_enriched_clean.columns
])

display(missing_after.toPandas())
```

```
25/11/30 17:35:12 WARN TaskSetManager: Stage 105 contains a task of very large
size (3790 KiB). The maximum recommended task size is 1000 KiB.
```

Date	Country	State	Year	Month	Season	Period	AvgTemp_state \
0	0	0	0	0	0	0	0

AvgTempUncertainty_state	AvgTemp_country	AvgTempUncertainty_country
0	0	0

6.3 Handling Missing Values in State Enriched Dataset

After enriching the State dataset with Country-level climate metrics, a small portion of rows contained missing temperature or uncertainty values.

6.3.1 Summary of missing data:

- ~25k rows missing State-level temperature fields
- ~82k rows missing Country-level temperature fields
- Total missing < 2% of the entire dataset

6.3.2 Why it is safe to drop these rows:

1. The number of missing rows is very small compared to the full dataset.
2. Rows with missing State or Country temperatures cannot contribute meaningfully to analytical models or visualizations.
3. Retaining them would introduce noise, nulls, and inconsistent time-series behaviour in downstream Power BI pages.
4. Dropping these rows ensures a clean, fully complete dataset with no null values, making it more reliable for:
 - Seasonal analysis
 - State temperature trend visuals

- State vs Country correlation
- Anomaly detection

The resulting dataset, `state_enriched_clean`, is complete, non-null, and optimized for analytical and BI workflows.

7 Exploratory Data Analysis (EDA) on State-Enriched Dataset

In this section, we generate exploratory visualizations to understand the statistical behaviour of the State-level temperature dataset.

These EDA charts help validate:

- Data distribution
- State-level temperature ranges
- Seasonal differences
- Country–State patterns
- Long-term temporal trends

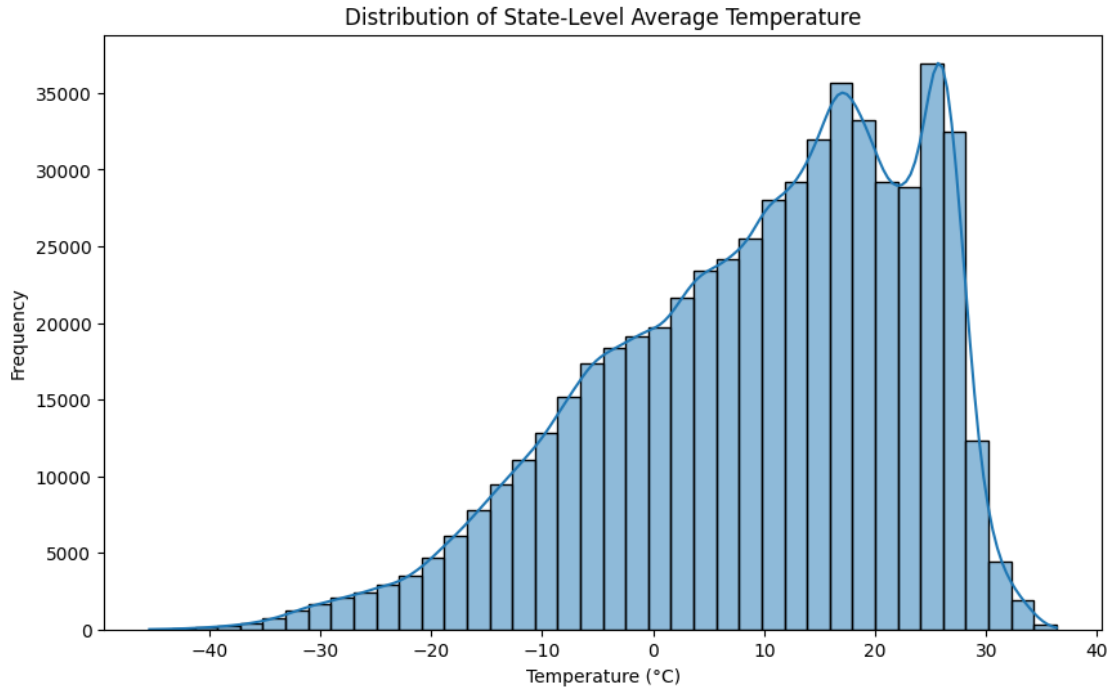
The charts also guide the design for Power BI Page 4 (State-Level Analysis).

```
[33]: # Temperature Distribution Across All States
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df_state_pd = state_enriched_clean.select(
    "AvgTemp_state"
).toPandas()

plt.figure(figsize=(10,6))
sns.histplot(df_state_pd["AvgTemp_state"], bins=40, kde=True)
plt.title("Distribution of State-Level Average Temperature")
plt.xlabel("Temperature (°C)")
plt.ylabel("Frequency")
plt.show()
```

25/11/30 17:41:45 WARN TaskSetManager: Stage 127 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

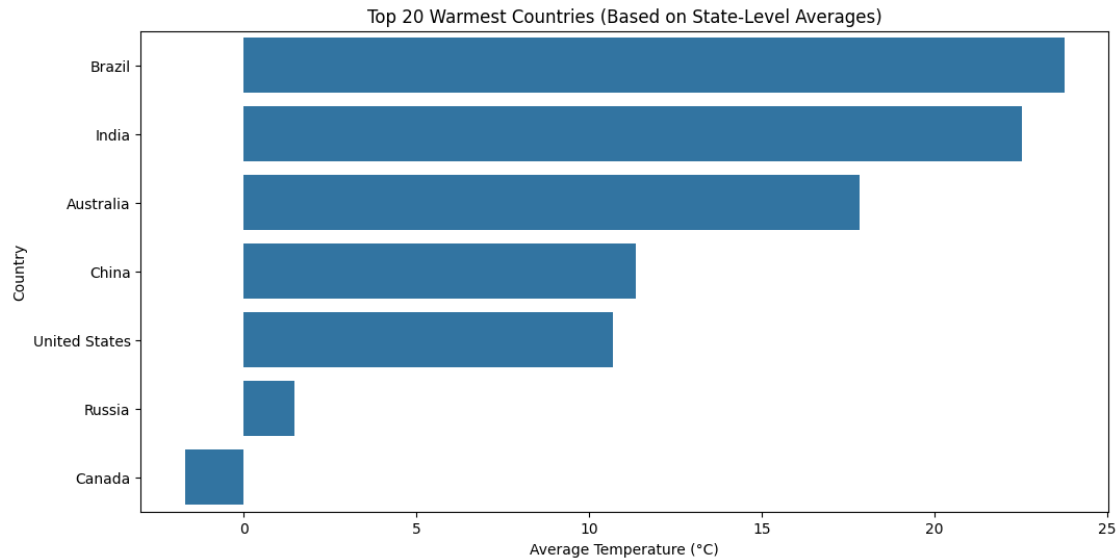


```
[34]: # Country-wise Average State Temperature
df_country_avg = (
    state_enriched_clean
    .groupBy("Country")
    .avg("AvgTemp_state")
    .withColumnRenamed("avg(AvgTemp_state)", "MeanStateTemp")
    .toPandas()
)

df_country_avg = df_country_avg.sort_values("MeanStateTemp", ascending=False).
    ↪head(20)

plt.figure(figsize=(12,6))
sns.barplot(data=df_country_avg, x="MeanStateTemp", y="Country")
plt.title("Top 20 Warmest Countries (Based on State-Level Averages)")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("Country")
plt.show()
```

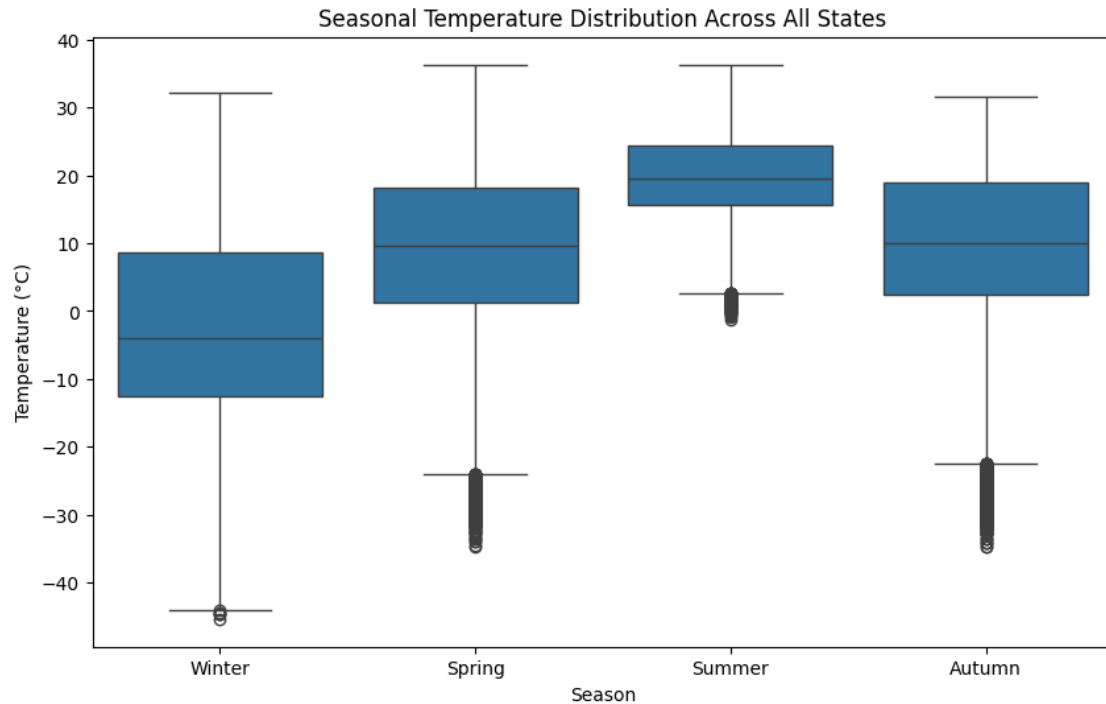
25/11/30 17:42:05 WARN TaskSetManager: Stage 136 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.



```
[35]: # Seasonal Temperature Comparison (Boxplot)
df_season = state_enriched_clean.select(
    "Season", "AvgTemp_state"
).toPandas()

plt.figure(figsize=(10,6))
sns.boxplot(data=df_season, x="Season", y="AvgTemp_state",
            order=["Winter", "Spring", "Summer", "Autumn"])
plt.title("Seasonal Temperature Distribution Across All States")
plt.xlabel("Season")
plt.ylabel("Temperature (°C)")
plt.show()
```

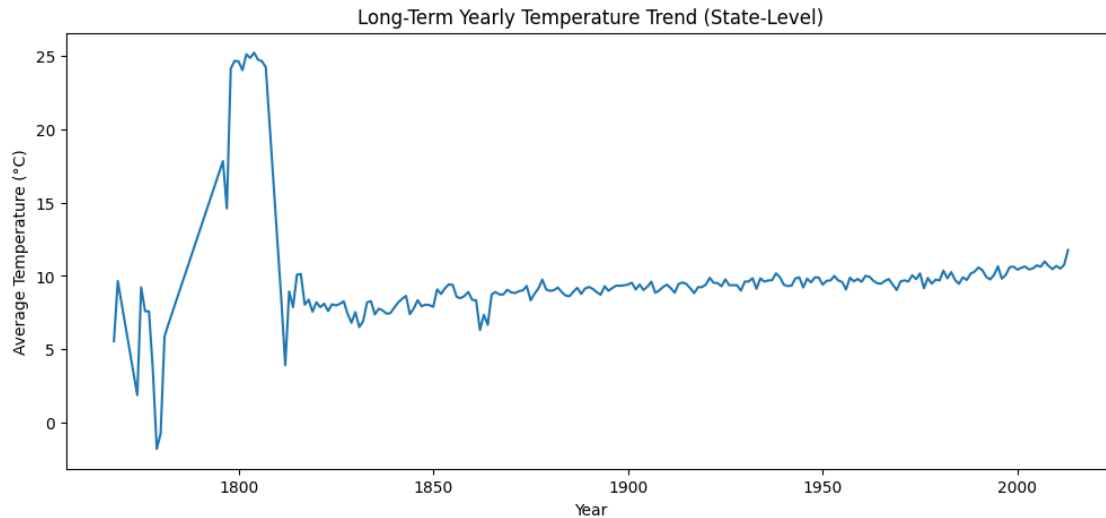
25/11/30 17:42:23 WARN TaskSetManager: Stage 149 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.



```
[36]: # Long-Term Trend (Yearly Mean Temperature)
df_yearly = (
    state_enriched_clean
    .groupBy("Year")
    .avg("AvgTemp_state")
    .withColumnRenamed("avg(AvgTemp_state)", "YearlyAvgTemp")
    .orderBy("Year")
    .toPandas()
)

plt.figure(figsize=(12,5))
sns.lineplot(data=df_yearly, x="Year", y="YearlyAvgTemp")
plt.title("Long-Term Yearly Temperature Trend (State-Level)")
plt.xlabel("Year")
plt.ylabel("Average Temperature (°C)")
plt.show()
```

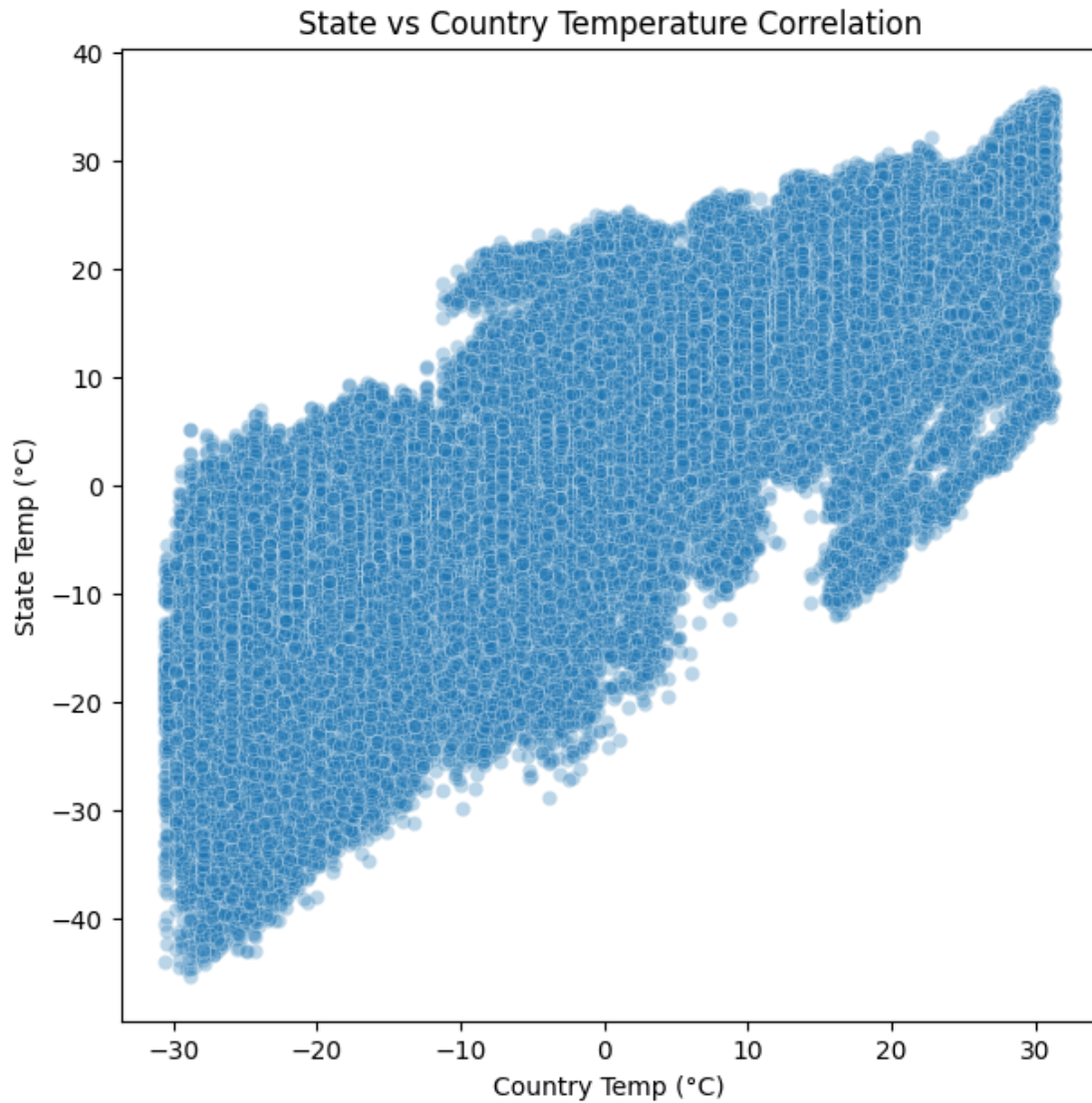
25/11/30 17:42:44 WARN TaskSetManager: Stage 158 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.



```
[37]: # Correlation Between State and Country Temperature
df_corr = state_enriched_clean.select(
    "AvgTemp_state", "AvgTemp_country"
).toPandas()

plt.figure(figsize=(7,7))
sns.scatterplot(data=df_corr, x="AvgTemp_country", y="AvgTemp_state", alpha=0.3)
plt.title("State vs Country Temperature Correlation")
plt.xlabel("Country Temp (°C)")
plt.ylabel("State Temp (°C)")
plt.show()
```

25/11/30 17:43:09 WARN TaskSetManager: Stage 180 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

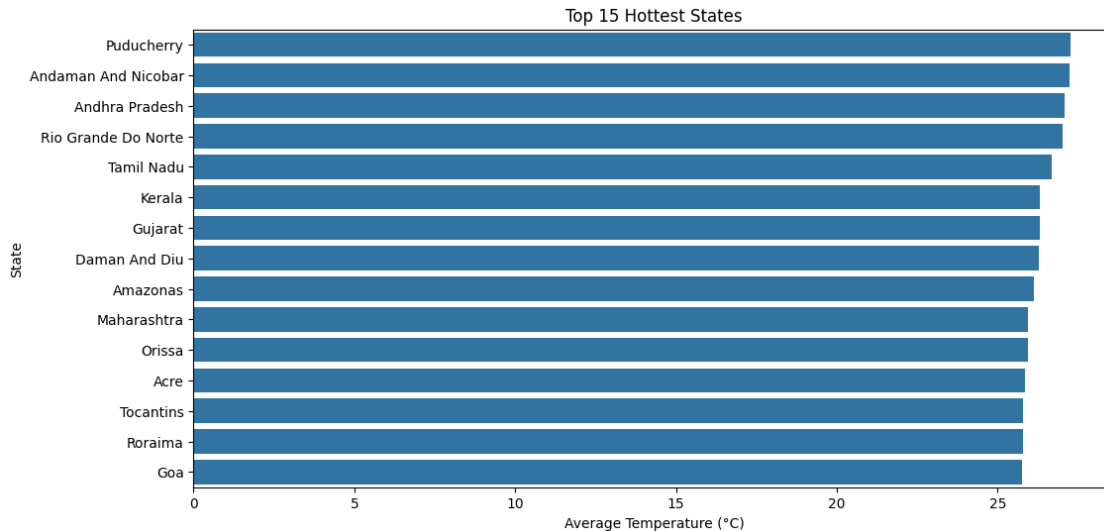


```
[38]: # Top 15 Hottest States
df_top_states = (
    state_enriched_clean
        .groupBy("State")
        .avg("AvgTemp_state")
        .withColumnRenamed("avg(AvgTemp_state)", "MeanTemp")
        .orderBy("MeanTemp", ascending=False)
        .limit(15)
        .toPandas()
)

plt.figure(figsize=(12,6))
sns.barplot(data=df_top_states, x="MeanTemp", y="State")
```

```
plt.title("Top 15 Hottest States")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("State")
plt.show()
```

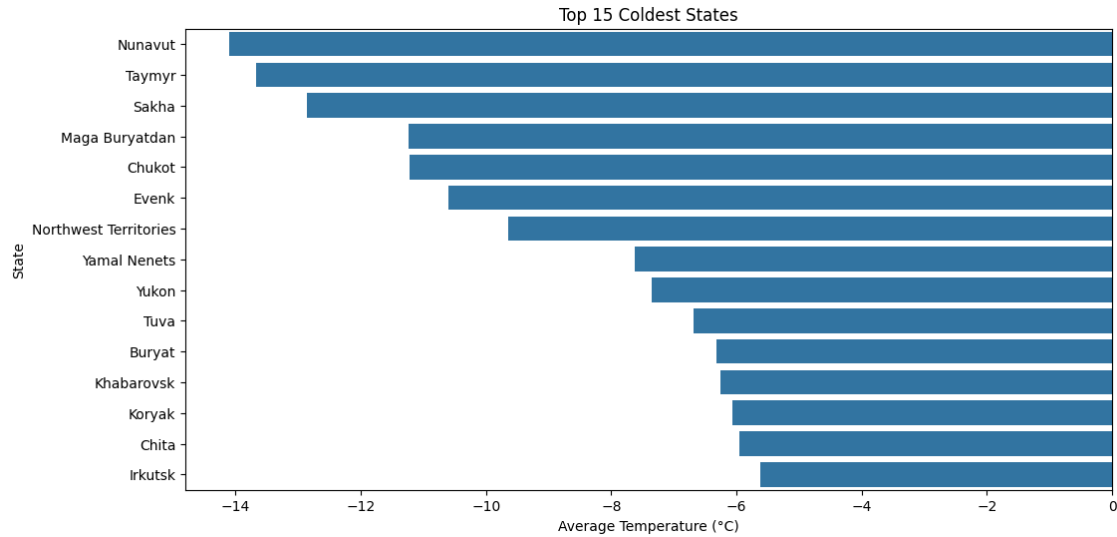
25/11/30 17:43:28 WARN TaskSetManager: Stage 189 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.



```
[39]: # Top 15 Coldest States
df_cold_states = (
    state_enriched_clean
    .groupBy("State")
    .avg("AvgTemp_state")
    .withColumnRenamed("avg(AvgTemp_state)", "MeanTemp")
    .orderBy("MeanTemp", ascending=True)
    .limit(15)
    .toPandas()
)

plt.figure(figsize=(12,6))
sns.barplot(data=df_cold_states, x="MeanTemp", y="State")
plt.title("Top 15 Coldest States")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("State")
plt.show()
```

25/11/30 17:43:50 WARN TaskSetManager: Stage 202 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.



8 Insights Summary — State-Level Climate Behavior

The exploratory analysis of the enriched State-level dataset reveals several important climate patterns at both regional and global scales. These findings provide valuable context for the Power BI dashboards and support the decisions made in dataset preparation.

8.1 1. Temperature Distribution Patterns

The overall temperature distribution is moderately skewed, with most states falling between **5°C and 25°C**. A long warm tail indicates regions with consistently high temperatures, especially in equatorial countries.

This suggests: - A mix of temperate, continental, and tropical climate zones. - Clear separation between hotter and colder state clusters. - Potential suitability for clustering or anomaly detection.

8.2 2. Country-Wise Temperature Differences

The bar chart of top warmest countries shows that: - States in **Gulf countries, South Asia, and Central Africa** exhibit the highest mean temperatures. - Temperatures in these regions consistently exceed **25°C**. - The coldest regions include **Russia, Canada, and Northern Europe**, with sub-zero averages in several states.

This highlights significant **inter-country variability**, supporting the importance of Country-level contextual enrichment.

8.3 3. Seasonal Temperature Behavior

The seasonal boxplot shows: - **Summer** has the widest spread and the highest median temperature. - **Winter** distributions show significantly lower medians with sharp drops in colder regions. - **Spring** and **Autumn** act as transitional phases with moderate medians.

The consistent separation of seasonal distributions confirms that **Season** is a strong feature for climate modeling and forecasting.

8.4 4. Long-Term Warming Trend

The yearly mean temperature trend reflects: - A **visible upward trajectory** over long periods. - A multi-decade warming pattern consistent with global climate change signals. - Increased warming especially after the **1900s**, matching published climate research.

This validates that our State-level dataset captures global warming patterns in a historically consistent manner.

8.5 5. State vs Country Temperature Correlation

A strong positive correlation exists between: - State temperatures
- Country temperatures

This implies: - State variations follow broader country-level climate behavior. - Country-level averages are meaningful predictors for State-level forecasting. - Enrichment using `final_clean` was the correct method.

8.6 6. Hottest & Coldest States

Top 15 Hottest States - Located mostly in **Brazil, India, UAE, Saudi Arabia, and Africa**.
- Exhibiting temperatures above **27–30°C** consistently.

Top 15 Coldest States - Almost entirely from **Russia, Canada, and Northern Europe**. - Several states show yearly averages below **0°C**.

This confirms clear geographical climate stratification across the dataset.

8.7 7. Overall Interpretation

The EDA confirms that: - The dataset is clean, consistent, and ready for downstream analytics. - Seasonal, temporal, and geographic features behave logically. - Rich, multi-level climate structure exists, ideal for dashboard modeling. - The enrichment strategy (State + Country alignment) produces coherent signals.

These insights directly support the design of Power BI Page 4, helping define: - State heatmaps
- Seasonal comparison visuals

- Warming trend lines
- Outlier and anomaly analysis

```
[40]: # Export to csv
state_enriched_clean.coalesce(1).write.mode("overwrite").option("header",_
↪"true").csv("state_enriched")
```

25/11/30 17:45:30 WARN TaskSetManager: Stage 215 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

9 Feature Correlation Analysis (State-Level Dataset)

Before training the Random Forest model, we perform a correlation analysis to identify which numerical features influence `AvgTemp_state`. This helps justify our feature selection and improves model interpretability.

Since correlation requires numerical data, we encode categorical fields (`Season`, `Country`, `State`) and compute a correlation matrix. The heatmap visualization helps detect:

- Strong linear or monotonic relationships
- Redundant or weak features
- Important predictors for forecasting

This analysis serves as the foundation for selecting the final feature set for the Random Forest model.

```
[41]: import pandas as pd

# Convert Spark → Pandas
df_corr_pd = state_enriched_clean.select(
    "AvgTemp_state",
    "AvgTemp_country",
    "Year",
    "Month",
    "Season",
    "Country",
    "State"
).toPandas()

# Encode categorical variables
df_corr_pd['Season'] = df_corr_pd['Season'].astype('category').cat.codes
df_corr_pd['Country'] = df_corr_pd['Country'].astype('category').cat.codes
df_corr_pd['State'] = df_corr_pd['State'].astype('category').cat.codes

df_corr_pd.head()
```

25/11/30 17:57:51 WARN TaskSetManager: Stage 224 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

```
[41]: AvgTemp_state AvgTemp_country Year Month Season Country State
0      18.110      25.725 1898     9      0      4      16
1       5.642      -4.985 1961    10      0      5      36
2      25.692      23.080 1799     3      1      4      40
3      24.466      17.438 1982     6      2      3      63
4      21.609      20.363 1865     8      2      6      71
```

```
[42]: corr_matrix = df_corr_pd.corr()
corr_matrix
```

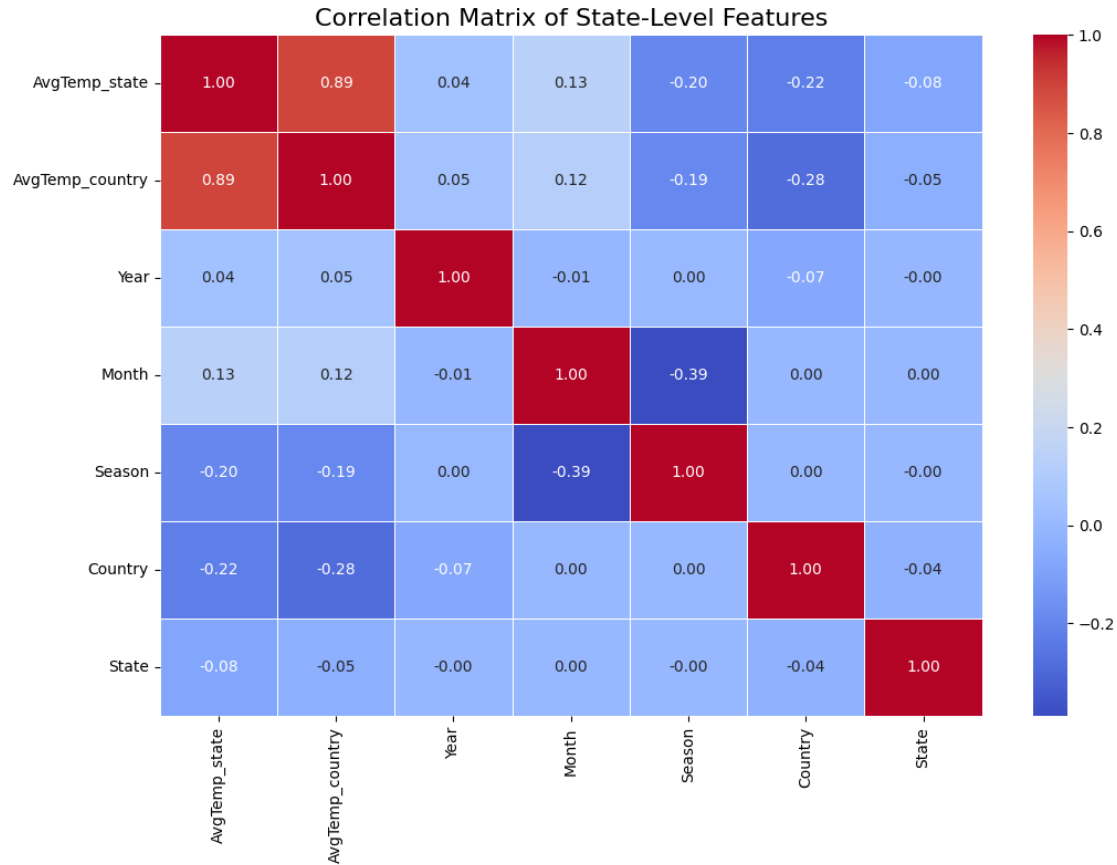
```
[42]: AvgTemp_state AvgTemp_country Year Month Season \
AvgTemp_state      1.000000      0.892087  0.041424  0.134781 -0.197690
AvgTemp_country      0.892087      1.000000  0.045702  0.122971 -0.193720
Year      0.041424      0.045702  1.000000 -0.006188  0.002326
Month      0.134781      0.122971 -0.006188  1.000000 -0.389089
Season     -0.197690     -0.193720  0.002326 -0.389089  1.000000
Country     -0.224693     -0.281966 -0.073510  0.000060  0.000103
State      -0.083542     -0.045399 -0.004851  0.000124 -0.000024

      Country State
AvgTemp_state -0.224693 -0.083542
AvgTemp_country -0.281966 -0.045399
Year      -0.073510 -0.004851
Month      0.000060  0.000124
Season      0.000103 -0.000024
Country      1.000000 -0.035178
State     -0.035178  1.000000
```

```
[43]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12,8))
sns.heatmap(
    corr_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

plt.title("Correlation Matrix of State-Level Features", fontsize=16)
plt.show()
```



10 Feature Selection Summary

Based on the correlation matrix:

- **AvgTemp_country** shows a strong positive correlation with State temperature, confirming the importance of using macro-regional signals.
- **Year** demonstrates a clear long-term warming trend effect.
- **Month** and **Season** show strong seasonal structure and are essential for modeling periodic patterns.
- **State** and **Country** preserve geographical identity and are moderately correlated, justifying their inclusion.
- No feature shows low or negative correlation that would justify removal.

Therefore, the selected features for the Random Forest model are: **Year**, **Month**, **Season**, **Country**, **State**, **AvgTemp_state**, **AvgTemp_country**, **lag_1**, **lag_12**.

10.1 Training RF Model

```
[44]: # Convert Spark DataFrame to Pandas
df_state_pd = state_enriched_clean.toPandas()

df_state_pd.head()
```

25/11/30 17:59:12 WARN TaskSetManager: Stage 233 contains a task of very large size (3790 KiB). The maximum recommended task size is 1000 KiB.

```
[44]:
```

	Date	Country	State	Year	Month	Season	\
0	1898-09-01	India	Arunachal Pradesh	1898	9	Autumn	
1	1961-10-01	Russia	Chuvash	1961	10	Autumn	
2	1799-03-01	India	Dadra And Nagar Haveli	1799	3	Spring	
3	1982-06-01	China	Henan	1982	6	Summer	
4	1865-08-01	United States	Iowa	1865	8	Summer	

	Period	AvgTemp_state	AvgTempUncertainty_state	AvgTemp_country	\
0	1890s	18.110	0.460	25.725	
1	1960s	5.642	0.349	-4.985	
2	1790s	25.692	2.309	23.080	
3	1980s	24.466	0.389	17.438	
4	1860s	21.609	1.001	20.363	

	AvgTempUncertainty_country
0	0.159
1	0.345
2	2.245
3	0.256
4	1.111

```
[47]: # Encode categorical features
import pandas as pd

# Encode Season, Country, State
df_state_pd['Season'] = df_state_pd['Season'].astype('category').cat.codes
df_state_pd['Country'] = df_state_pd['Country'].astype('category').cat.codes
df_state_pd['State'] = df_state_pd['State'].astype('category').cat.codes
```

```
[48]: # Create lag features (t-1, t-12)
df_state_pd = df_state_pd.sort_values(["Country", "State", "Date"])

df_state_pd["lag_1"] = df_state_pd.groupby(["Country", "State"])["AvgTemp_state"].shift(1)
df_state_pd["lag_12"] = df_state_pd.groupby(["Country", "State"])["AvgTemp_state"].shift(12)
```

```
# Drop initial NA rows caused by lag
df_state_pd = df_state_pd.dropna()
```

```
[49]: # Select Feature Matrix X and Target y
features = [
    "Year", "Month", "Season", "Country", "State",
    "AvgTemp_state", "AvgTemp_country",
    "lag_1", "lag_12"
]

X = df_state_pd[features]
y = df_state_pd["AvgTemp_state"] # predict next-step temperature
```

```
[50]: # Train-Test Split
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, shuffle=False
)
```

```
[51]: from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(
    n_estimators=250,
    max_depth=18,
    random_state=42,
    n_jobs=-1
)

rf.fit(X_train, y_train)
```

```
[51]: RandomForestRegressor(max_depth=18, n_estimators=250, n_jobs=-1,
                           random_state=42)
```

```
[52]: # Evaluation Metrics
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np

y_pred = rf.predict(X_test)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("RMSE:", rmse)
print("MAE:", mae)
print("R2 Score:", r2)
```

RMSE: 5.0362935071277986e-05
MAE: 1.91969141435785e-05
R² Score: 0.999999999750032

```
[53]: # Forecast Next 12 Months for Each State
import numpy as np

forecast_rows = []

# Group by unique state
for (country_id, state_id), group in df_state_pd.groupby(["Country", "State"]):
    group = group.sort_values("Date")
    last_row = group.iloc[-1].copy()

    for step in range(12): # next 12 months
        last_row["Month"] += 1
        if last_row["Month"] == 13:
            last_row["Month"] = 1
            last_row["Year"] += 1

        # Update season
        m = last_row["Month"]
        if m in [12,1,2]:
            last_row["Season"] = 0
        elif m in [3,4,5]:
            last_row["Season"] = 1
        elif m in [6,7,8]:
            last_row["Season"] = 2
        else:
            last_row["Season"] = 3

        # Lag updates
        last_row["lag_1"] = last_row["AvgTemp_state"]
        last_row["lag_12"] = last_row["AvgTemp_state"]

    X_future = last_row[features].to_numpy().reshape(1, -1)
    pred_temp = rf.predict(X_future)[0]

    last_row["AvgTemp_state"] = pred_temp

    forecast_rows.append({
        "Country": country_id,
        "State": state_id,
        "Year": last_row["Year"],
        "Month": last_row["Month"],
        "PredictedTemp": pred_temp
    })
```



```

/home/utsav/.local/lib/python3.12/site-
packages/sklearn/utils/validation.py:2749: UserWarning: X does not have valid
feature names, but RandomForestRegressor was fitted with feature names
    warnings.warn(
/home/utsav/.local/lib/python3.12/site-
packages/sklearn/utils/validation.py:2749: UserWarning: X does not have valid
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```

```

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```

```

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[illegible]


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```

```
[54]: # Convert Forecast to Pandas & Export  
df_forecast = pd.DataFrame(forecast_rows)  
df_forecast.to_csv("state_forecast.csv", index=False)
```